

Don't be squared ..., Pulp Fiction, Quintine Tarantino, 1994

<https://www.youtube.com/watch?v=IY48IXSK4nk>



Logical Level | Relational Data Model

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A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model Level

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

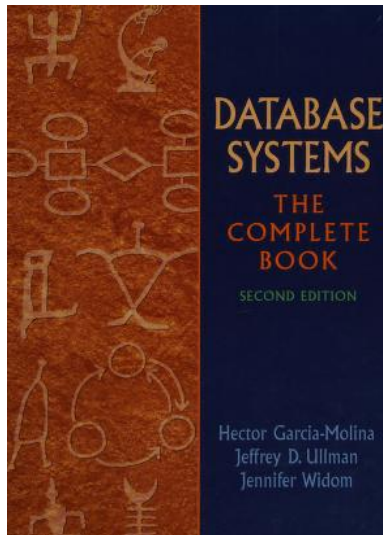
Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

Outline: W04A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation

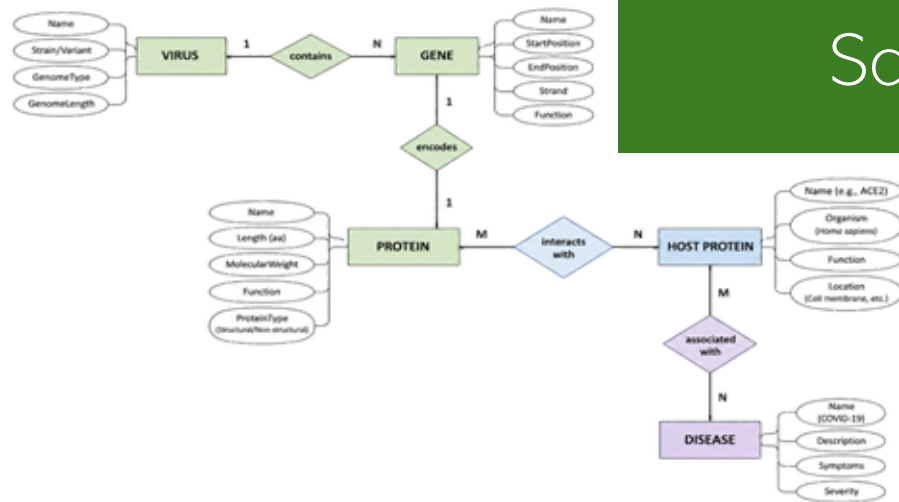
IIII) IS-A to Relation

Hands-On

4

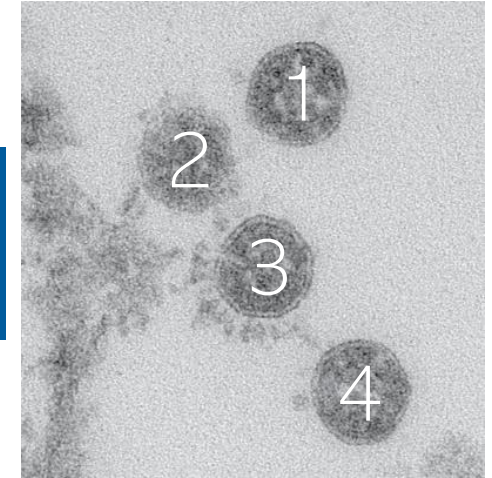
Abstractions: Logical Representation

How **instances** of entities, **values** of their attributes, their relationships should be represented based on a schema.



Schema

Real-World
Instances of Entity



Storage

Abstractions: Logical Representation

Common logical models include:

~1960: Object Oriented

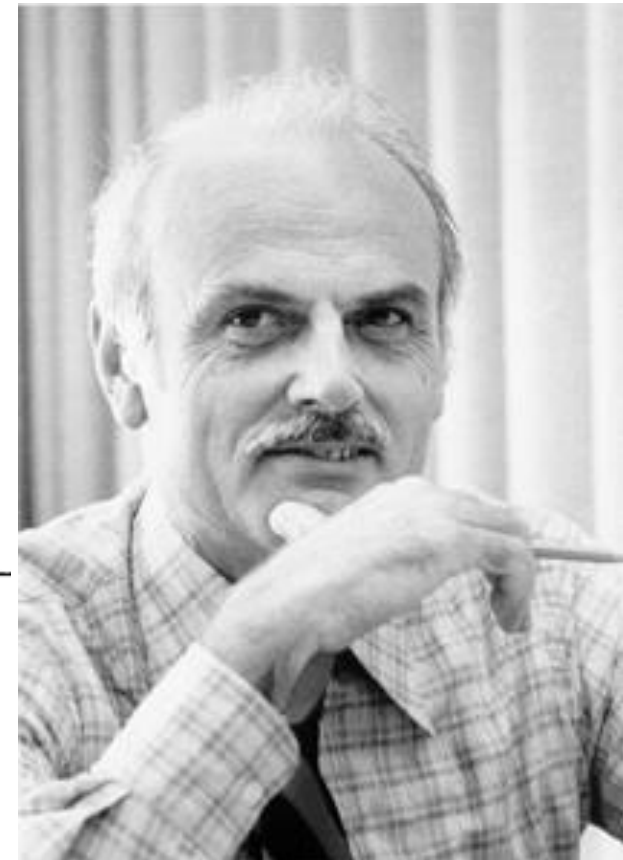
~1969: Relational: Mathematical Relation

~1996: XML

Edgar Frank "Ted" Codd, IBM, 1969, 1970
Information Retrieval

**A Relational Model of Data for
Large Shared Data Banks**

E. F. CODD
IBM Research Laboratory, San Jose, California



6 Logical Model: Relational

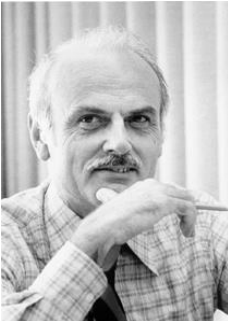
An Entity Set with Attributes coming from domains $A, B, C, \dots$, is like,

$$A \times B \times C \times \dots = \{(a, b, c, \dots) \mid a \in A, b \in B, c \in C, \dots\}$$

that can include all possible entities of the Entity Set.

→ Data instance is then represented as a **Tuple**: $(a, b, c, \dots)$

→ Entity Set is then a **Relation**: $\{(a, b, c, \dots), (a', b', c', \dots), \dots\}$



Relational Model: Formal vs. Informal Terminology

Relation

Schema	<u>Title</u>	ReleaseDate	RunningTime	Genre
Tuple 1	2001: A Space Odyssey	English	1 42	SciFi
Tuple 2	Rosemary's Baby	English	1 36	Horror
Tuple 3	The Birds	English	1 19	Horror
...	...	...	...	

Relation vs. Table

Tuple vs. Row

Domain vs. Column

8

Relational Model: Everything is Relation

Entity Set $\rightarrow$ Relation

Relationship Set $\rightarrow$ Relation

Data Modeling
in
RDBMS

Real World Entity

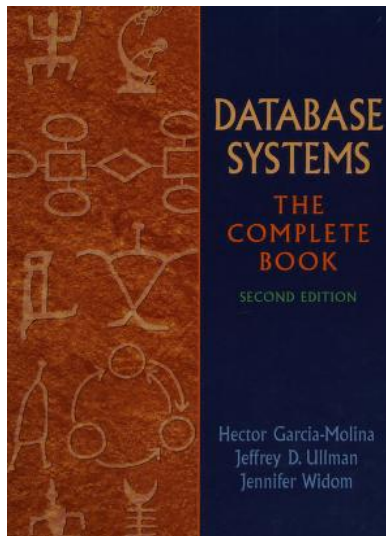
Conceptual Level | Entity Set & Relationship Set

Conceptual Level | Logical Level | Relation

Conceptual Level | Logical Level | Physical Level | SQL

Conceptual Level | Logical Level | Computable Entity

Outline: W04A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

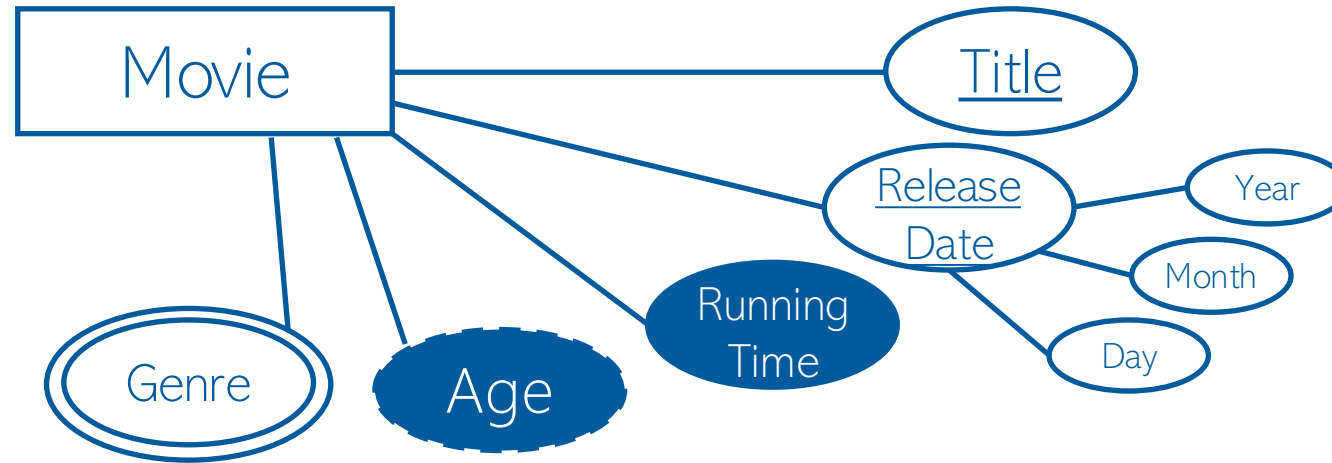
III) Relationship to Relation

IIII) IS-A to Relation

Hands-On

10

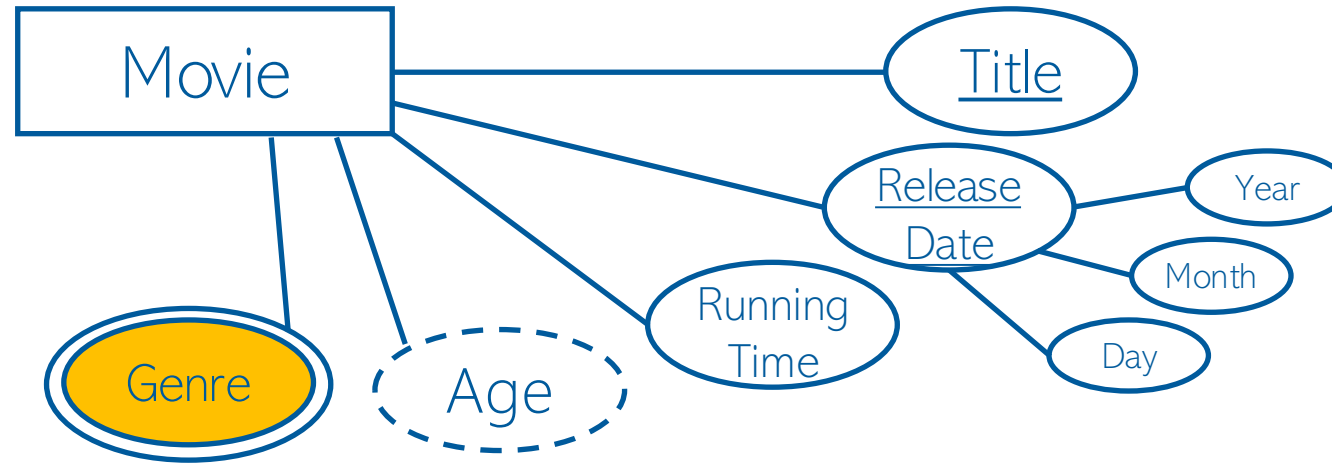
Relational Model: ER to Relation Transformation



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

11

Relational Model: ER to Relation Transformation



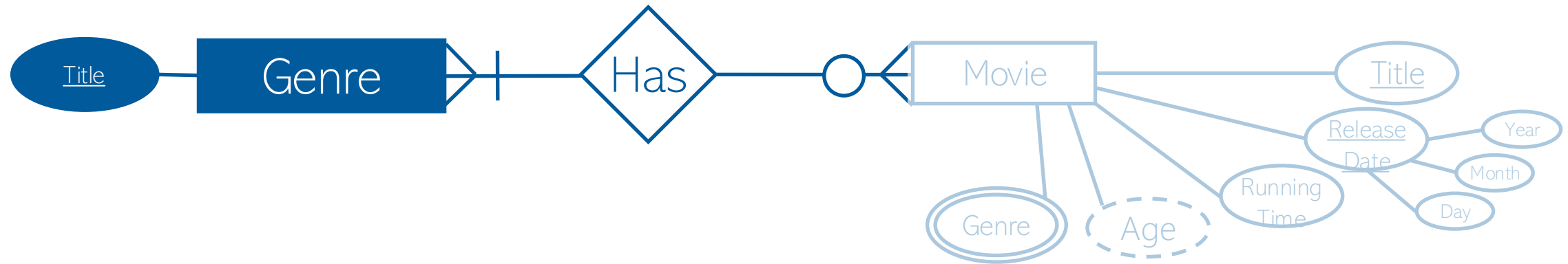
Relational model does not have multivalued attributes!

R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)



13

Relational Model: ER to Relation Transformation

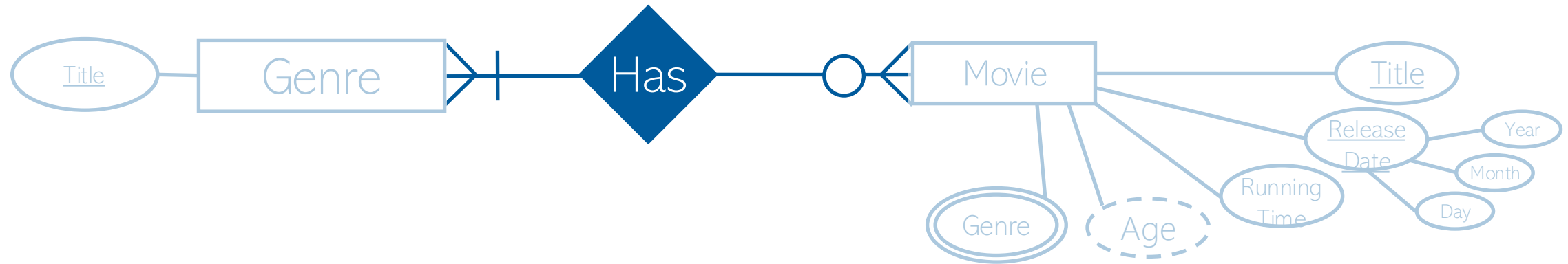


R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

14

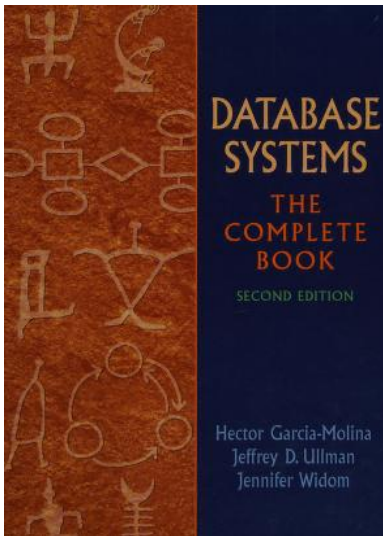
Relational Model: ER to Relation Transformation



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Genre(Title)

Outline: W04B



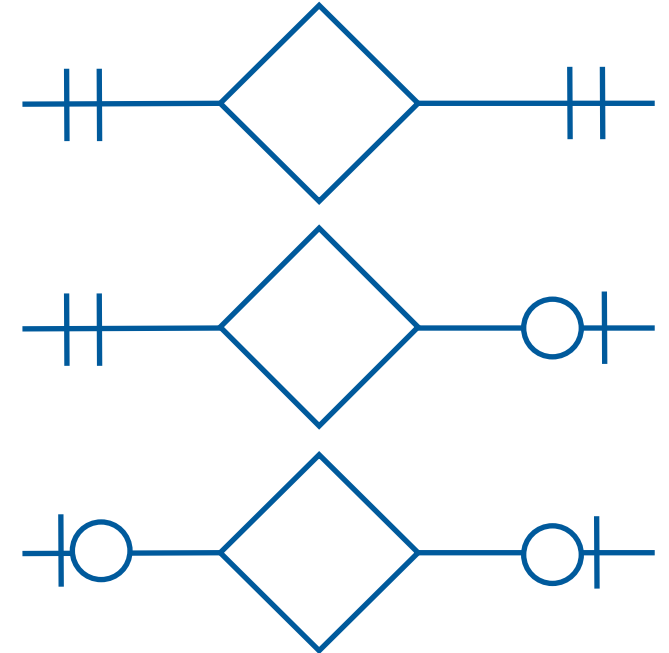
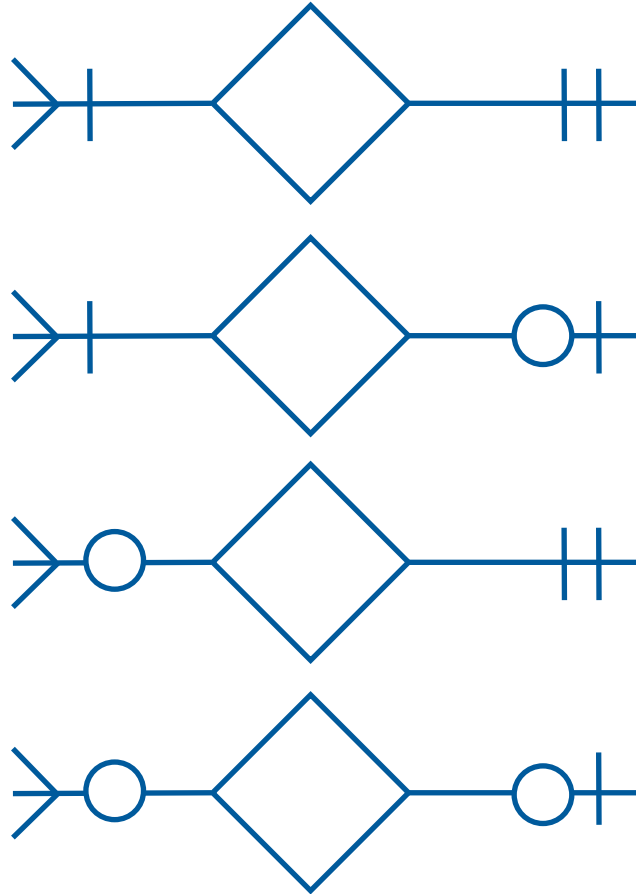
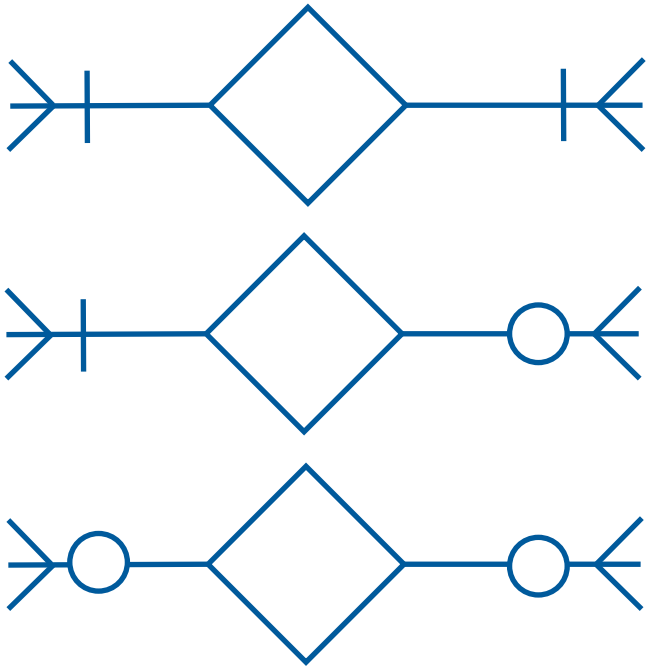
CH02 (2nd Ed.)

- I) Relational Modeling
- II) Entity to Relation
- III) Relationship to Relation
- IIII) IS-A to Relation

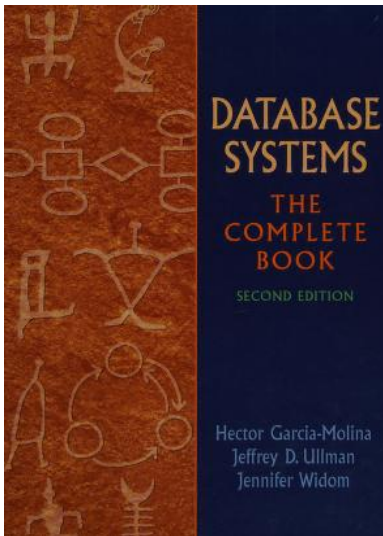
Hands-On

16

Relationship



Outline: W04B



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

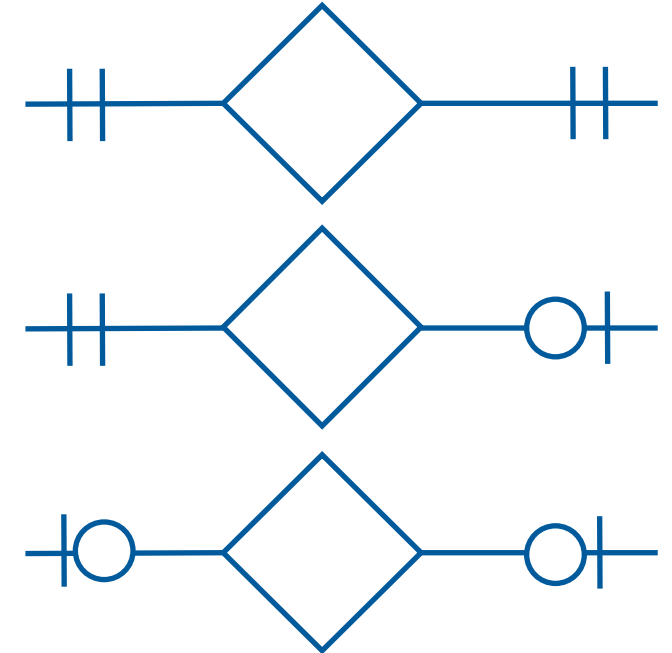
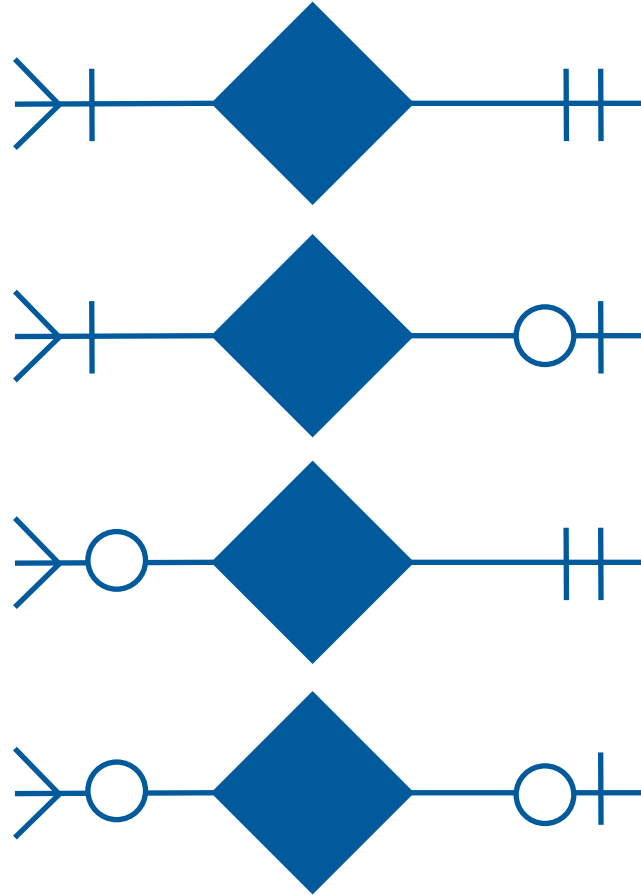
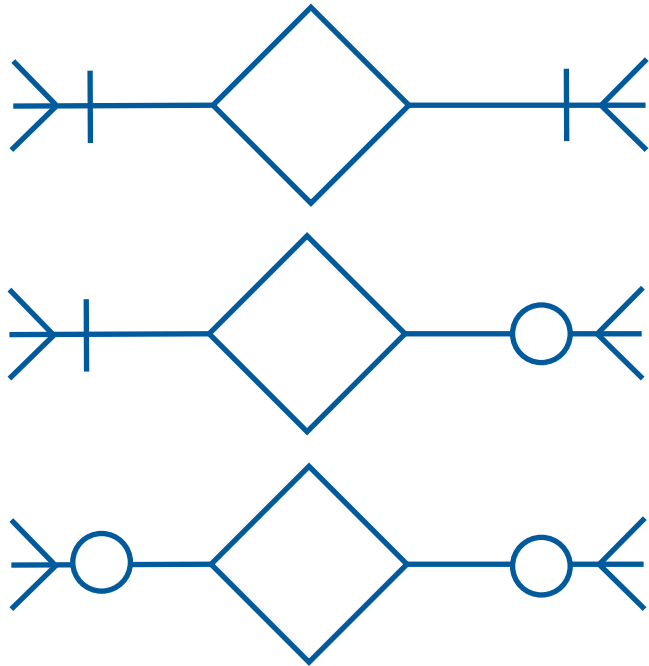
III) Relationship to Relation (Many-One)

IIII) IS-A to Relation

Hands-On

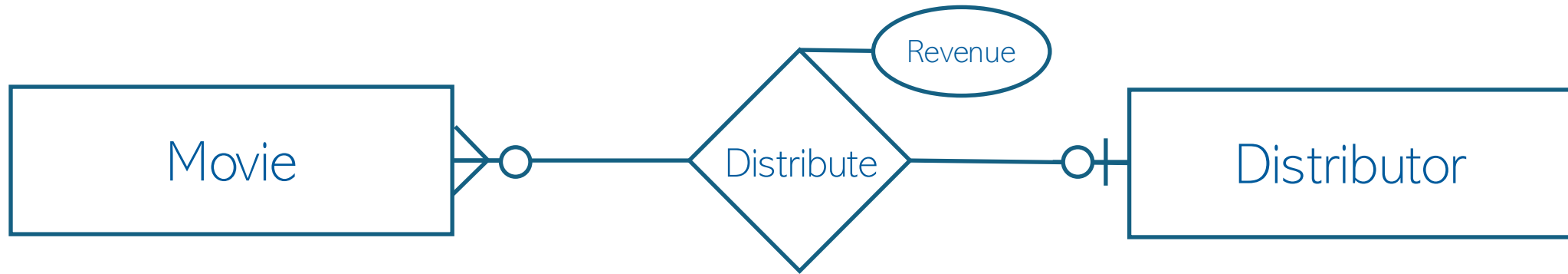
18

Relationship: Many-One (One-Many)



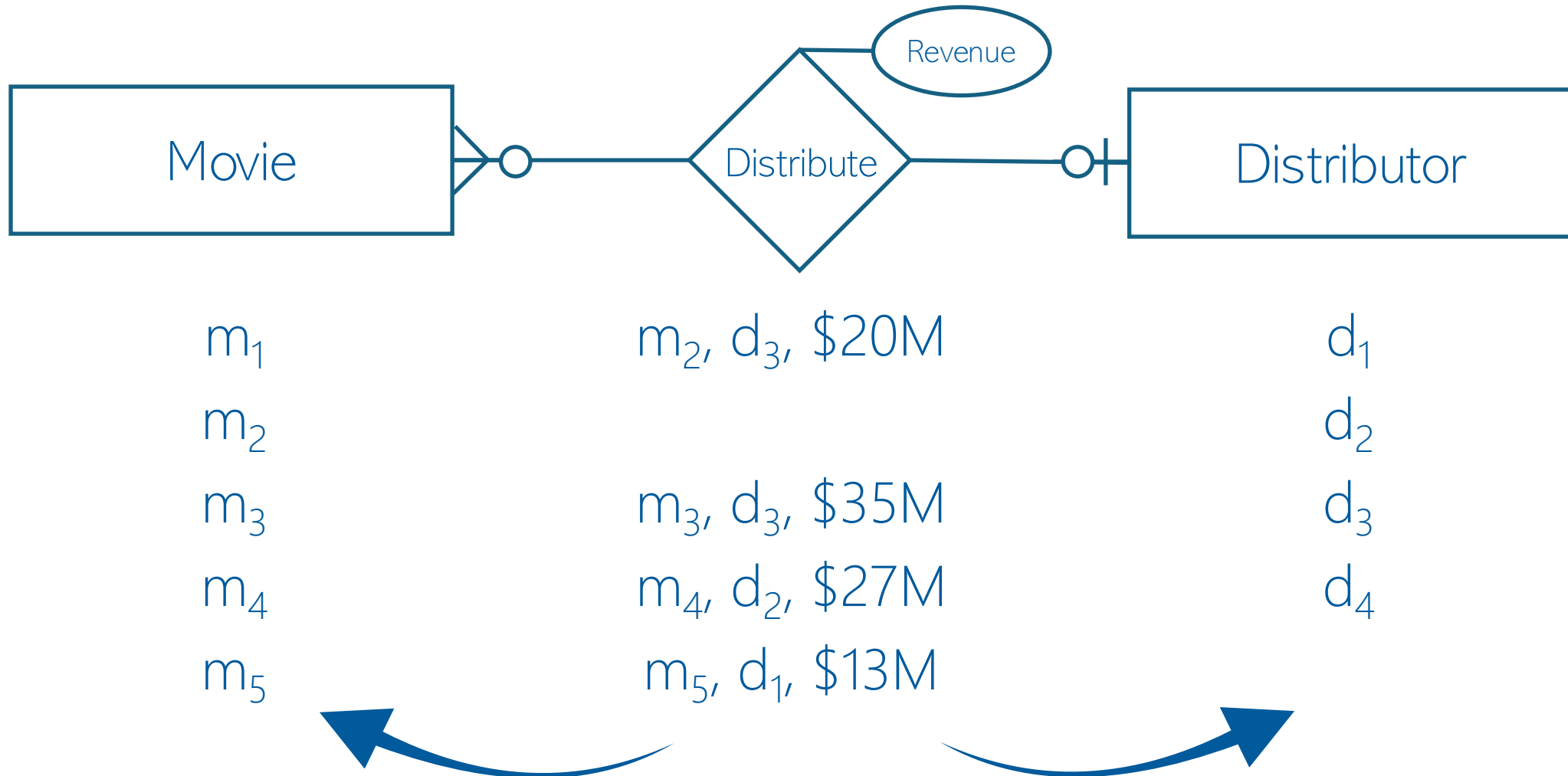
19

Relationship: Many-One (One-Many)

 m_1 $m_2, d_3, \$20M$ d_1 m_2 ~~$m_2, d_4, \$28M$~~ d_2 m_3 $m_3, d_3, \$35M$ d_3 m_4 $m_4, d_2, \$27M$ d_4 m_5 $m_5, d_1, \$13M$

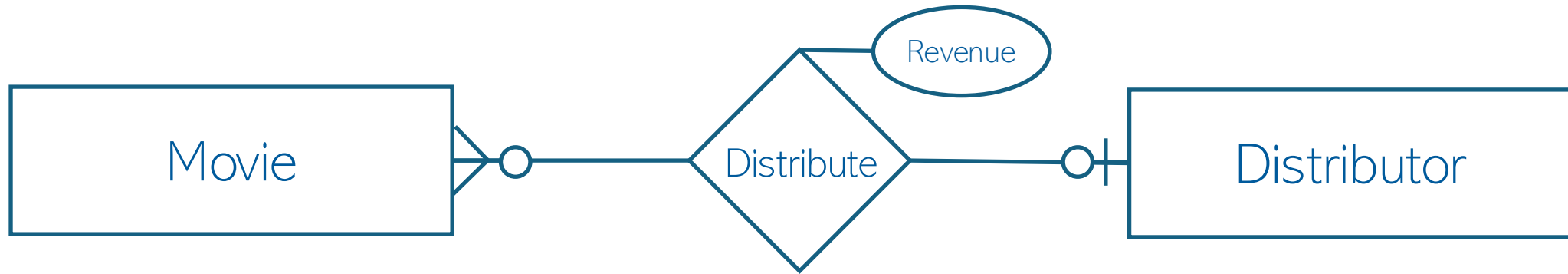
20

Relationship: Many-One (One-Many)



21

Relationship: Many-One (One-Many)

 m_1 m_2 m_3 m_4 m_5

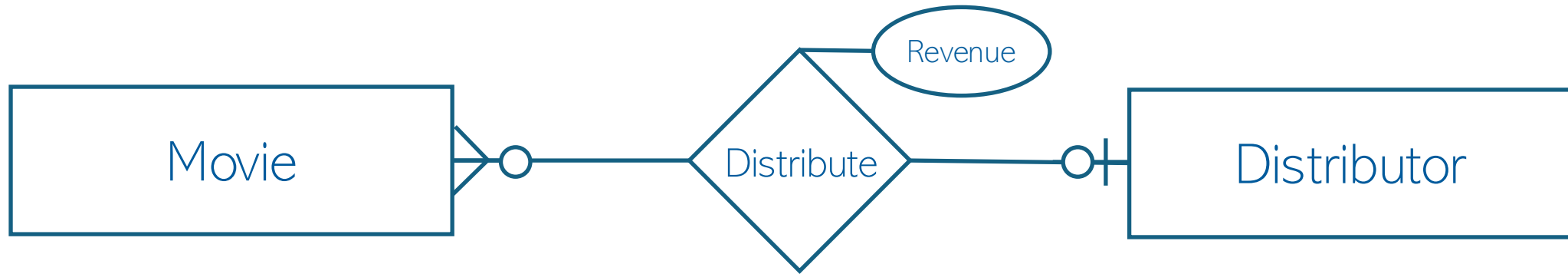
A distributor needs new attributes
for a new movie to distribute!

Inserting data instance needs
changes to schema!

 $d_1, m_5, \$13M$ $d_2, m_4, \$27M$ $d_3, m_2, \$20M, m_3, \$35M$ d_4 

22

Relationship: Many-One (One-Many)



NULL, NULL, m_1

d_3 , \$20M, m_2

d_3 , \$35M, m_3

d_2 , \$27M, m_4

d_1 , \$13M, m_5

As a movie can have maximum 1 distributor, if any, the number of attributes to add is always fixed.

d_1

d_2

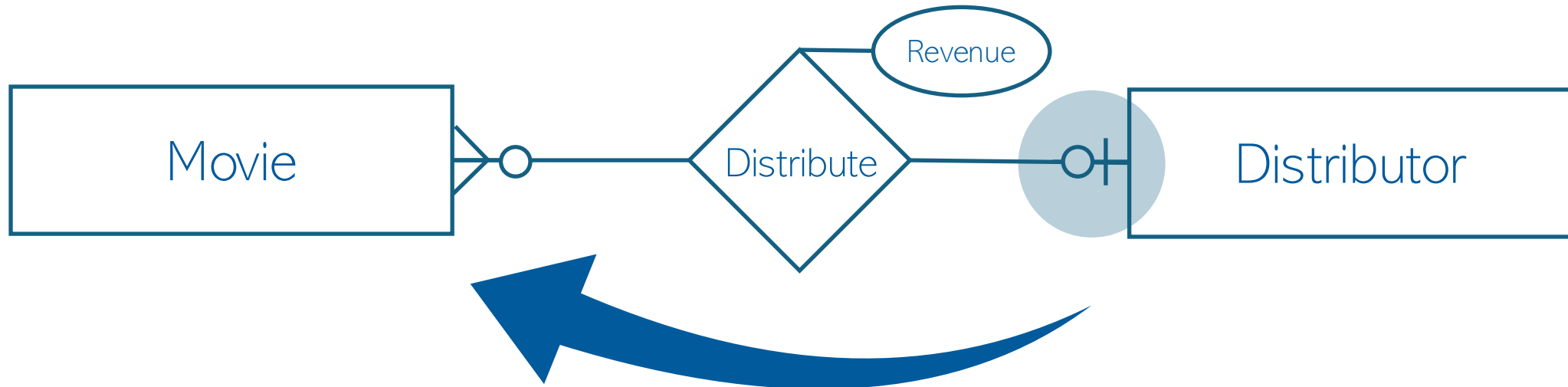
d_3

d_4



23

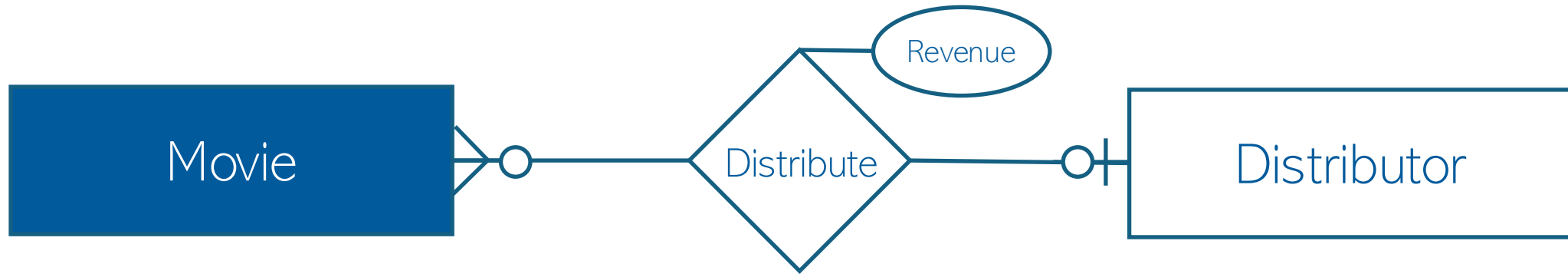
Relationship: Many-One (One-Many)



Everything goes to entity set with cardinality one (i.e., visually many side)
Because it only needs to store one relationship at max.

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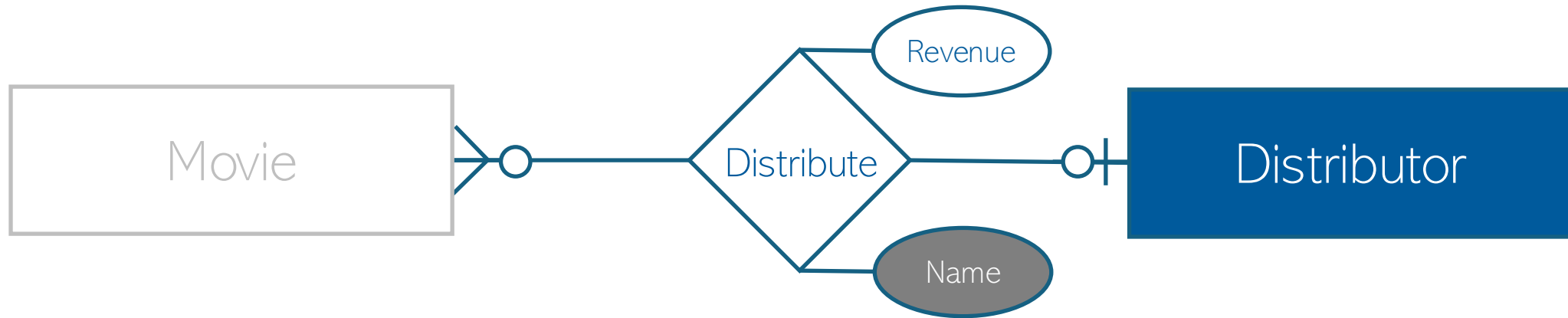
Relationship: Many-One (One-Many)



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

25

Relationship: Many-One (One-Many)

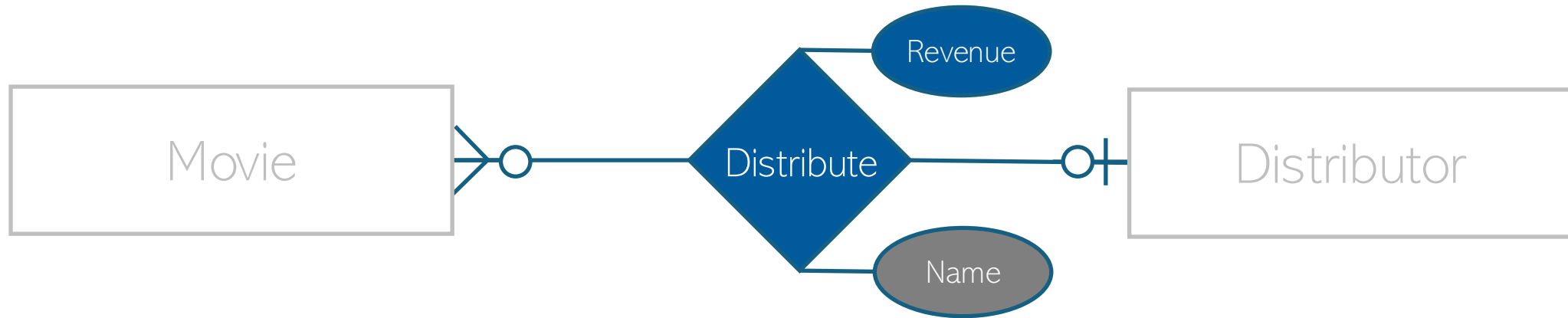


R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

R2: Distributor(Name, Address, POBox, Website, ...)

26

Relationship: Many-One (One-Many)

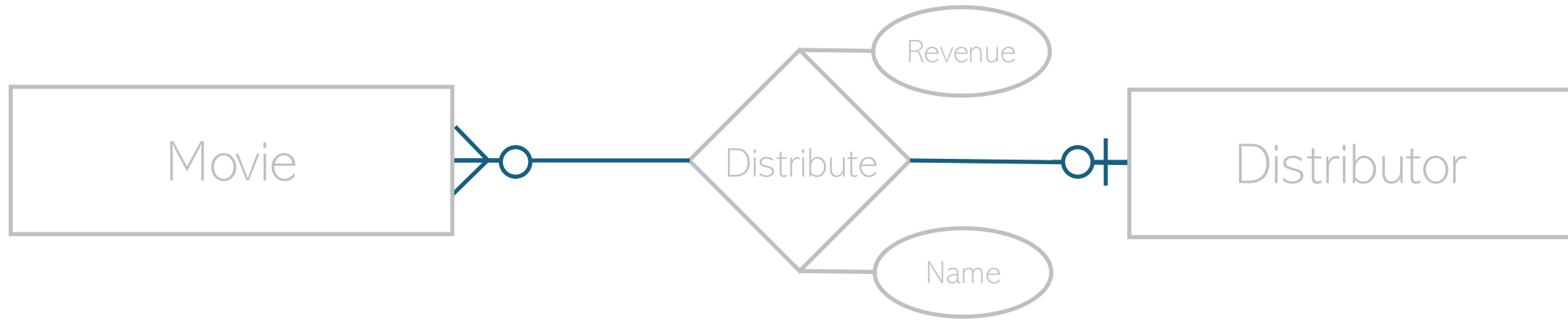


R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, **Distributor.Name**, Revenue)

R2: Distributor(Name, Address, POBox, Website, ...)

27

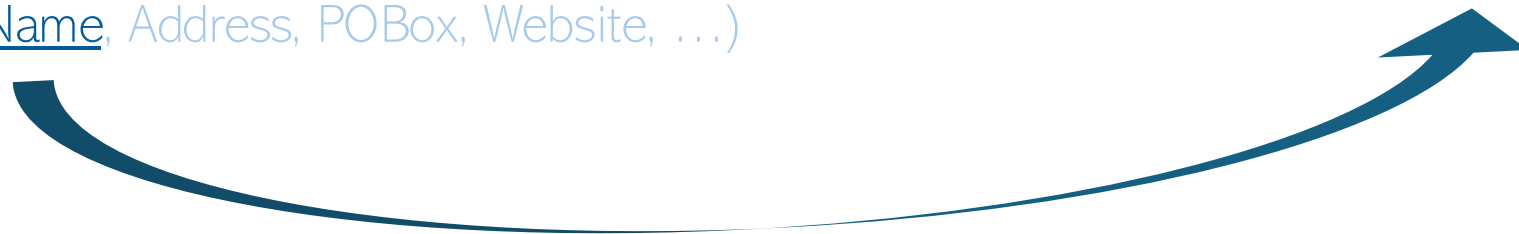
Relationship: Many-One (One-Many)



Primary Key (PK) from other relation: Foreign Key (FK)

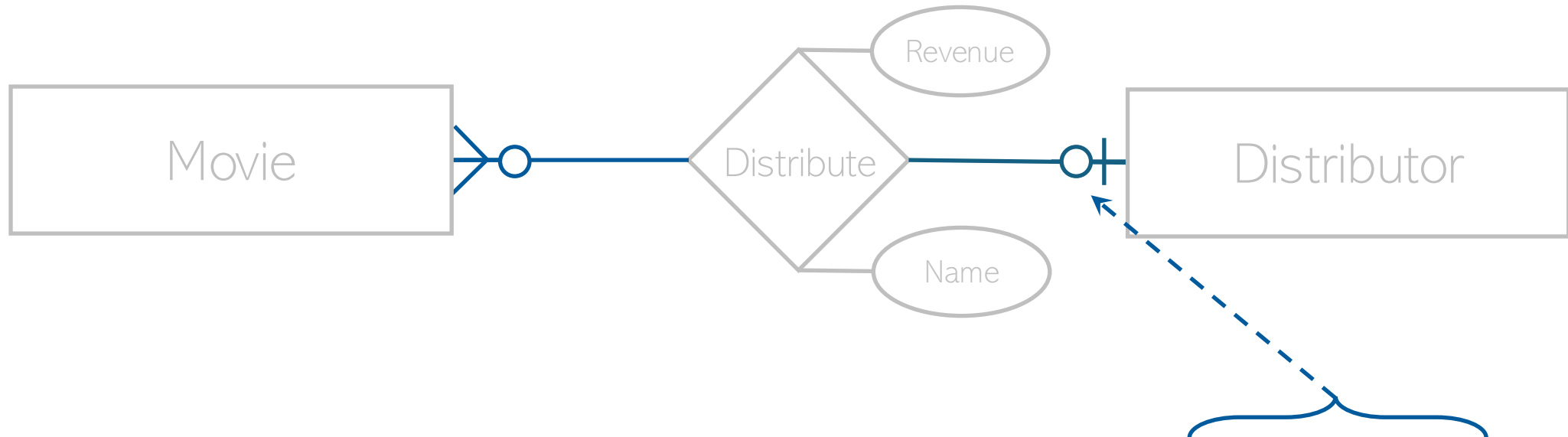
R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, **Distributor**.Name, Revenue)

R2: Distributor(Name, Address, POBox, Website, ...)



28

Relationship: Many-One (One-Many)



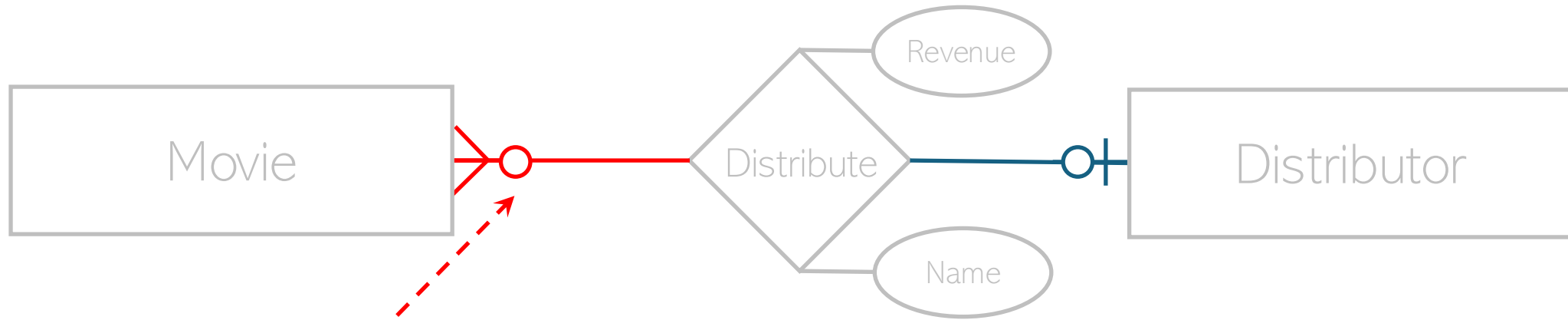
R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, **Distributor.Name**, Revenue)

R2: Distributor(Name, Address, POBox, Website, ...)

FK: Optional (Nullable)

29

Relationship: Many-One (One-Many)



R1: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, Distributor.Name, Revenue)

R2: Distributor(Name, Address, POBox, Website, ...)

FK: Optional (Nullable)

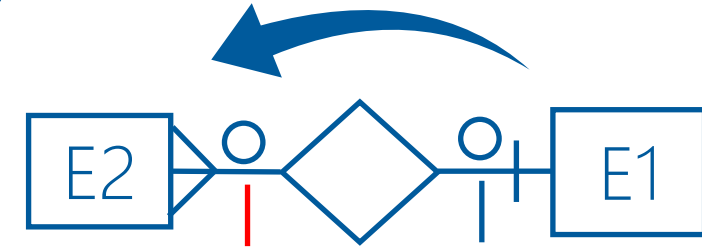
Optional: No need to convert/check. Why?

Mandatory: We need to enforce it, but relational falls short!

30

Relationship: Many-One (One-Many)

Input: Many-One relationship between E2 and E1, i.e.,
Output: Relations R1 for E1 and R2 for E2.

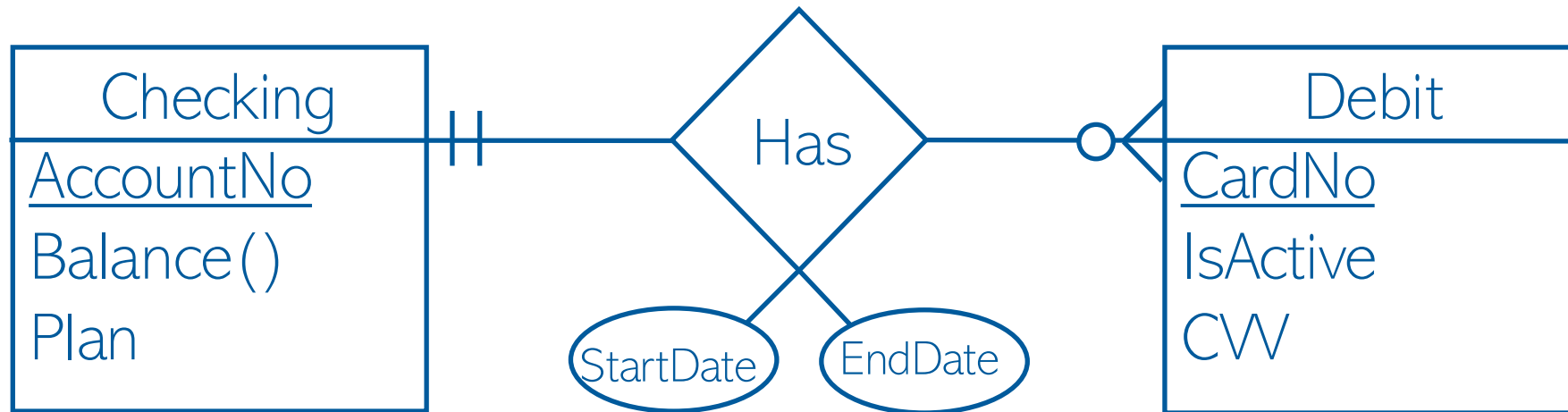
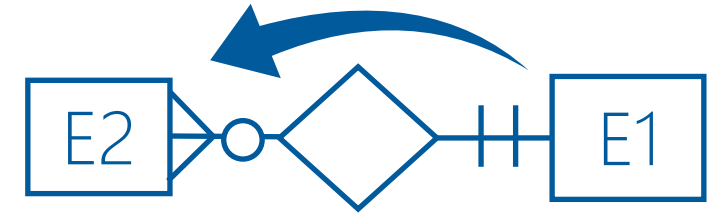


1. For E1, create relation R1 with the same attributes and keys as in E1
2. For E2, create relation R2 with the same attributes and keys as in E2
3. **[Foreign Key Set]** Add key set of E1 to R2
4. Add attributes of relationship set to R2
5. If E2 ordinality is optional then make foreign key set optional
else make foreign key set mandatory

Herein, we do not care about E1's ordinality.

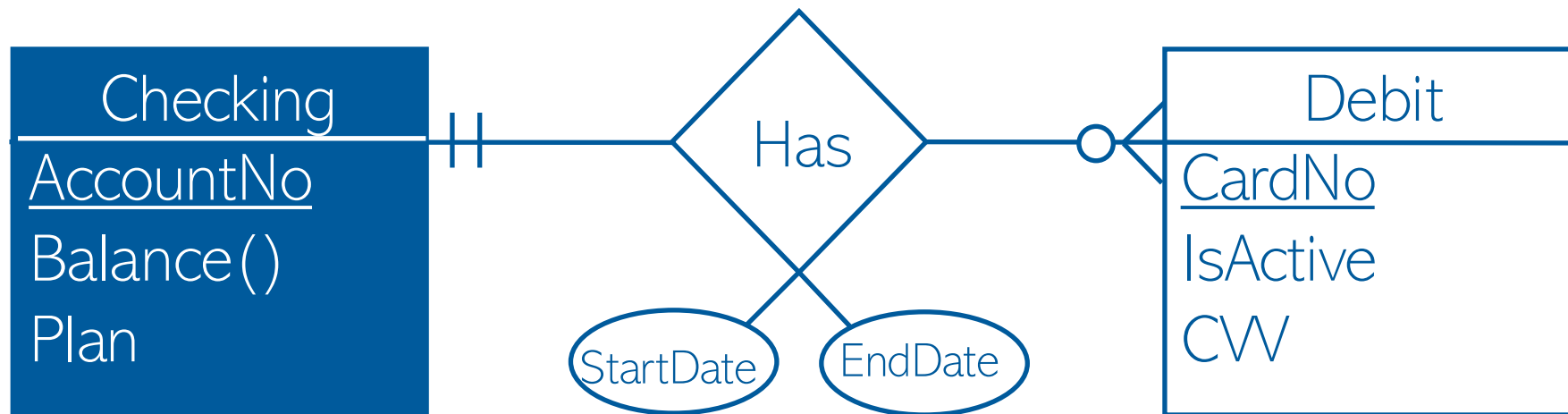
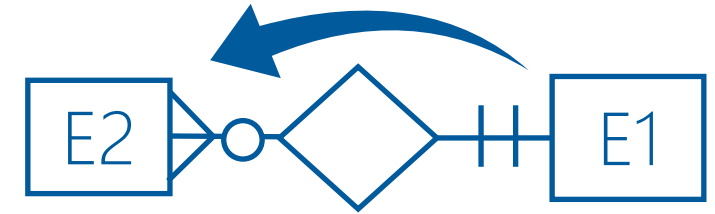
31

Relationship: Many-One (One-Many)



32

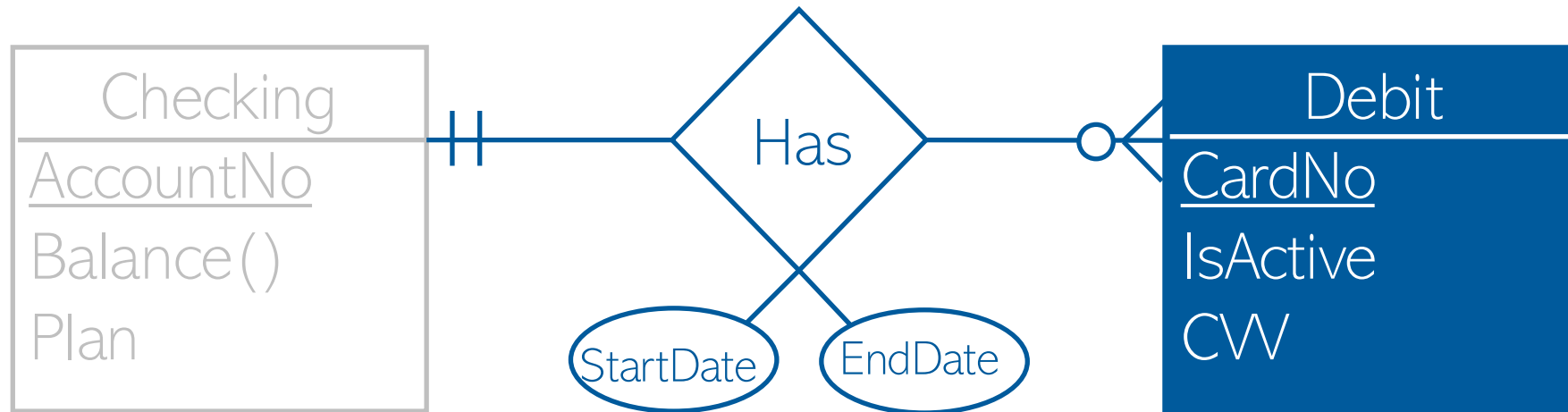
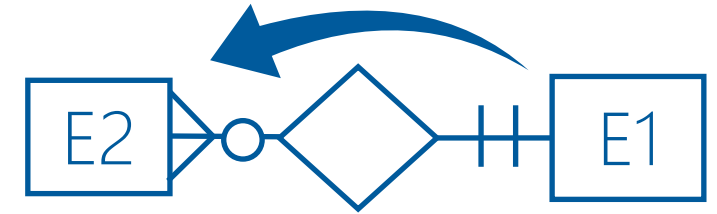
Relationship: Many-One (One-Many)



R1: Checking(AccountNo, Balance, Plan)

33

Relationship: Many-One (One-Many)

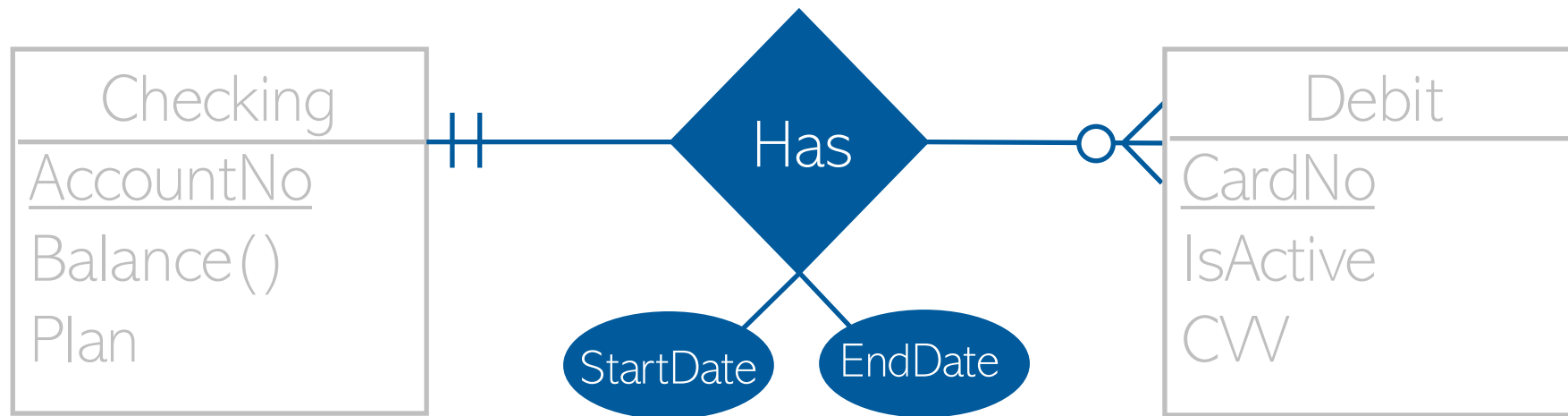
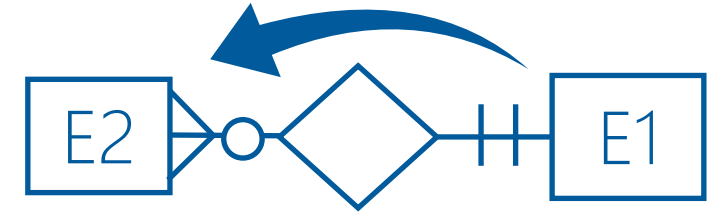


R1: Checking(AccountNo, Balance, Plan)

R2: Debit(CardNo, IsActive, CVV)

34

Relationship: Many-One (One-Many)

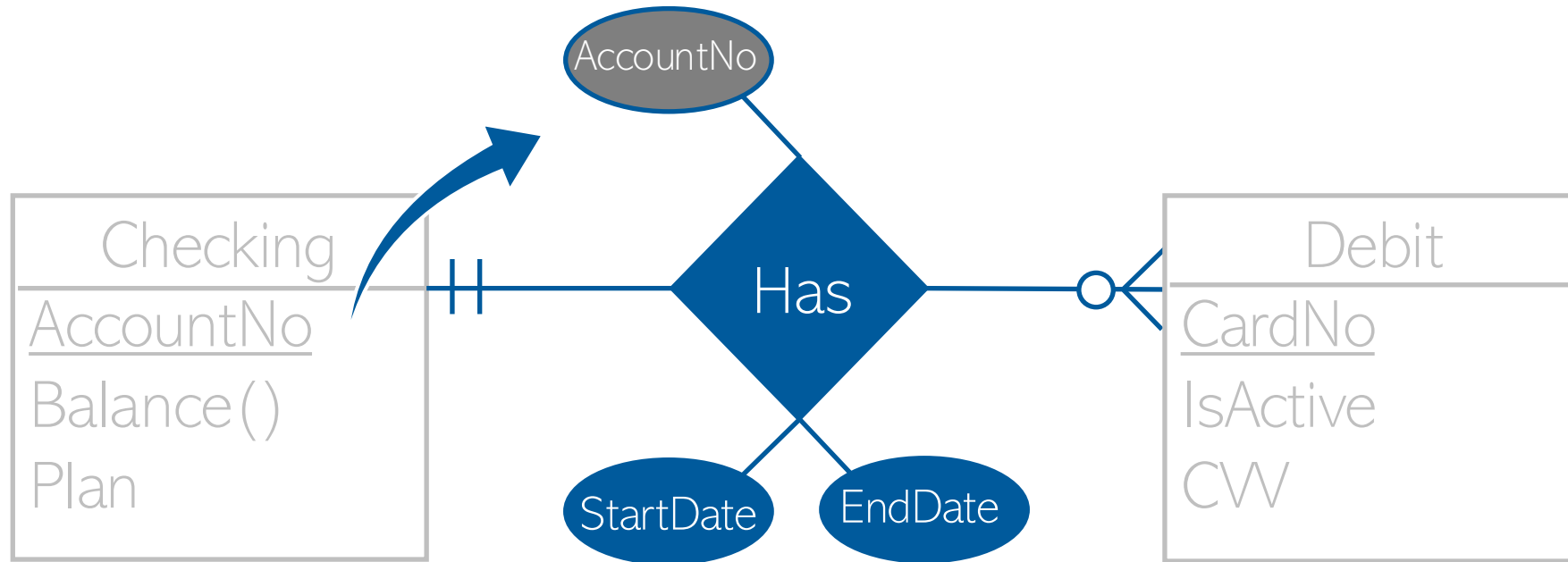
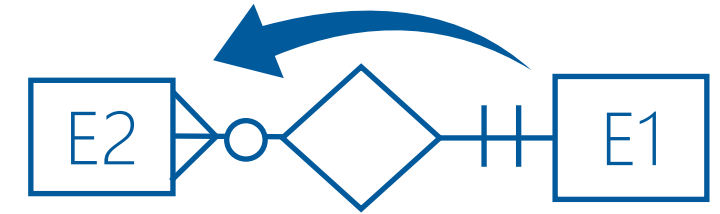


R1: Checking(AccountNo, Balance, Plan) ← ?

R2: Debit(CardNo, IsActive, CVV) ← ?

35

Relationship: Many-One (One-Many)



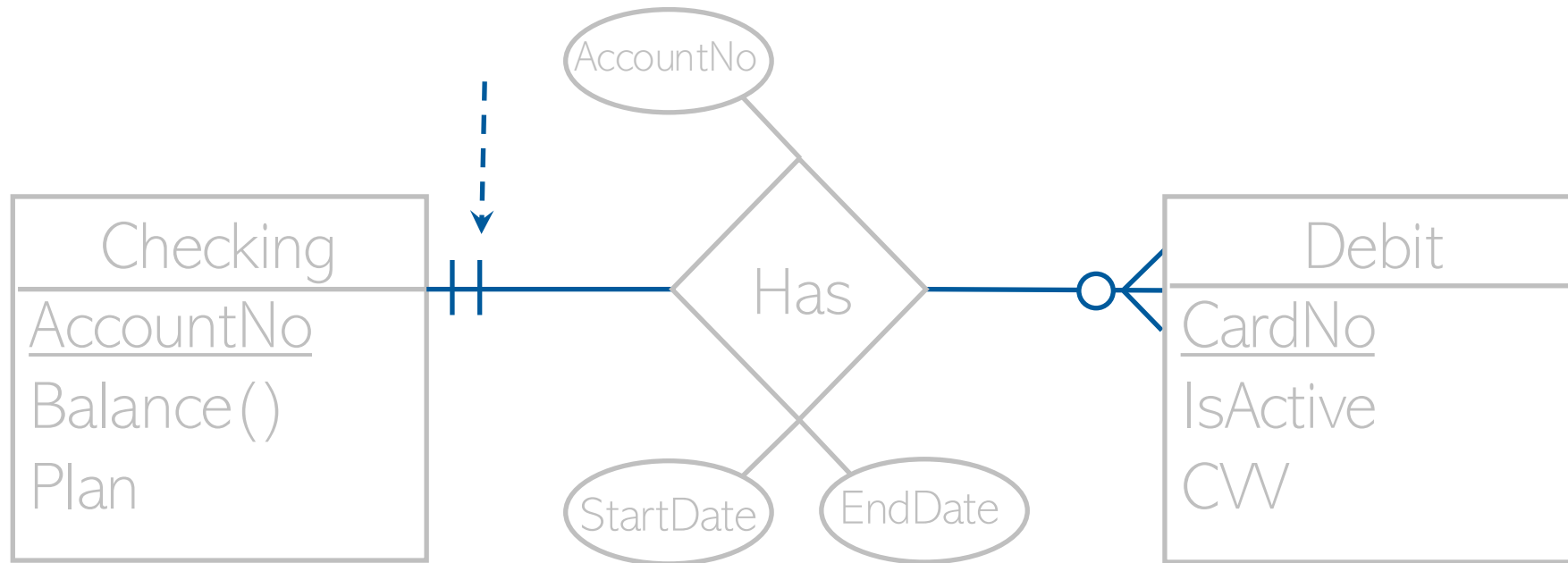
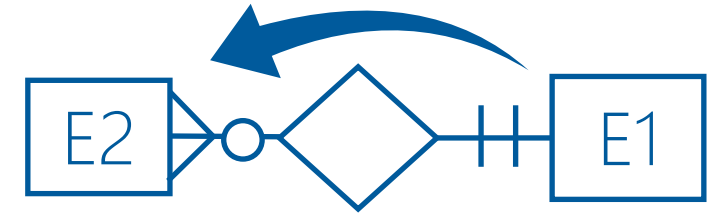
R1: Checking(AccountNo, Balance, Plan)

R2: Debit(CardNo, IsActive, CVV, Checking.AccountNo, StartDate, EndDate)

FK

36

Relationship: Many-One (One-Many)



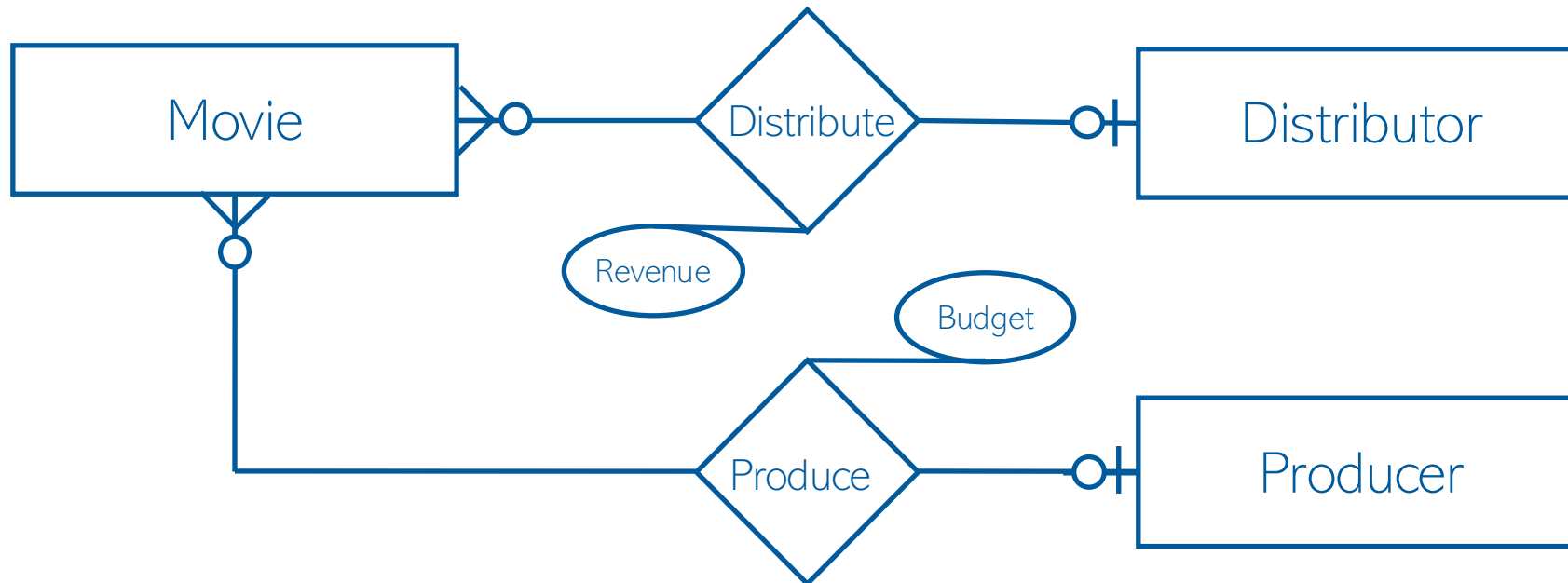
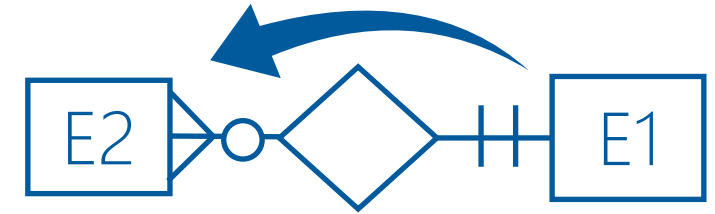
R1: Checking(AccountNo, Balance, Plan)

R2: Debit(CardNo, IsActive, CVV, Checking.AccountNo, StartDate, EndDate)

FK, Mandatory

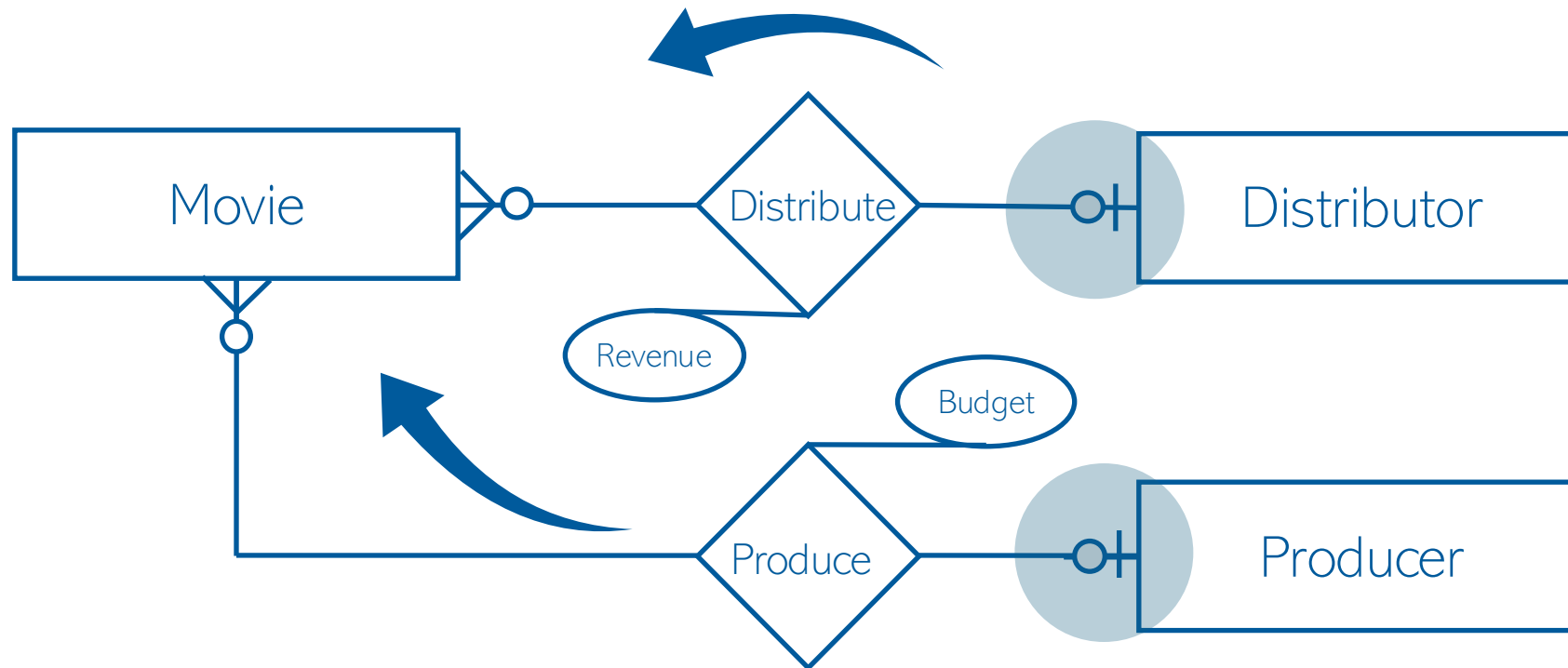
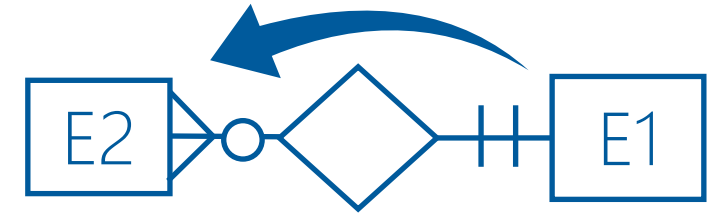
37

Relationship: Multiple Many-One



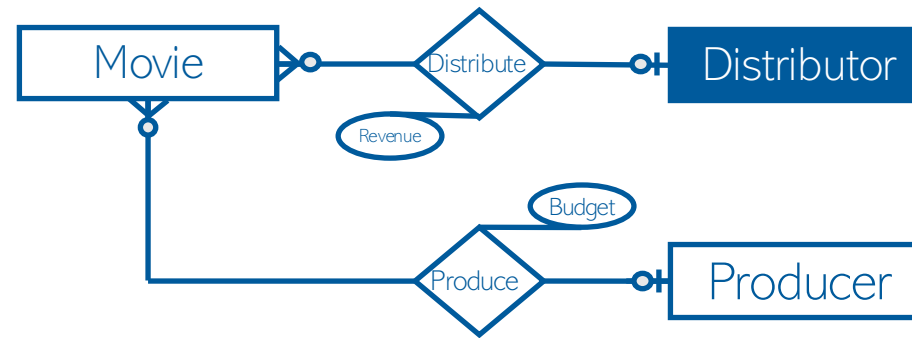
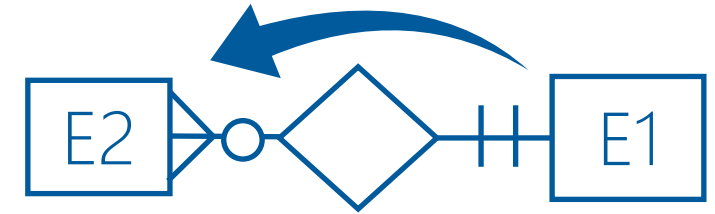
38

Relationship: Multiple Many-One



39

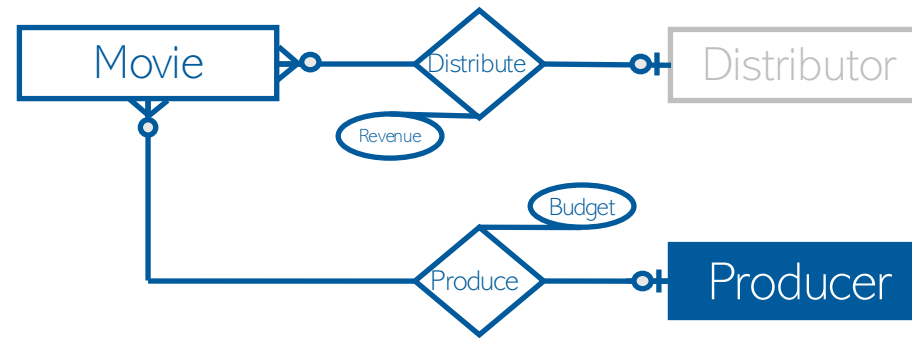
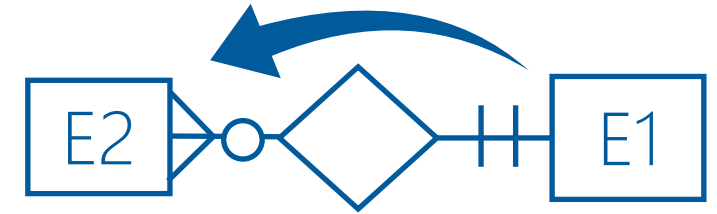
Relationship: Multiple Many-One



R1: Distributor(Name, Address, Phone, ...)

40

Relationship: Multiple Many-One

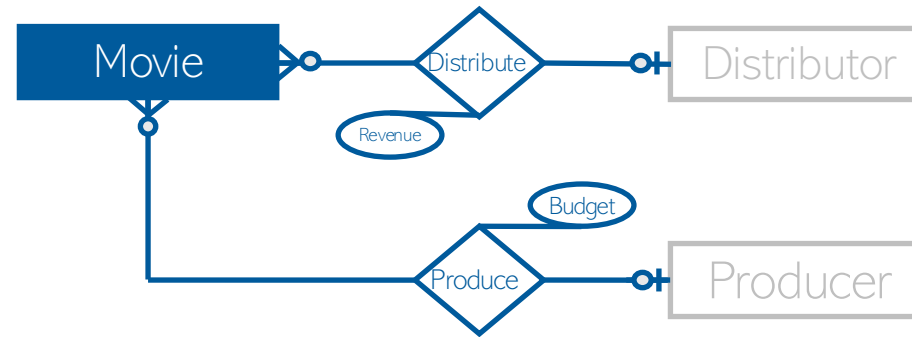
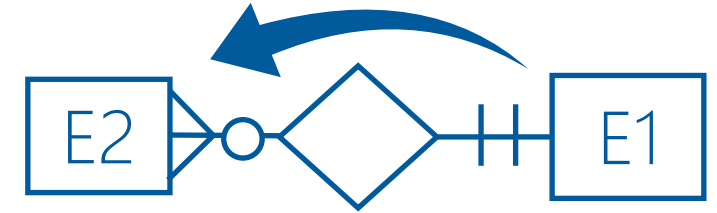


R1: Distributor(Name, Address, Phone, ...)

R2: Producer(Name, Studio, Address, Phone, ...)

41

Relationship: Multiple Many-One



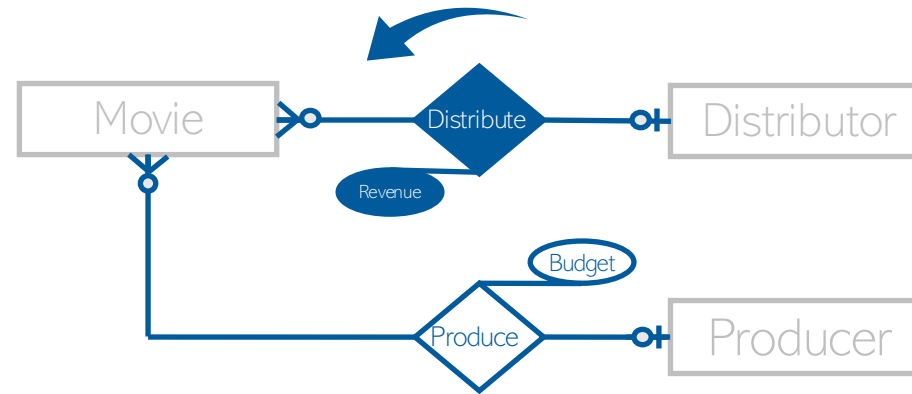
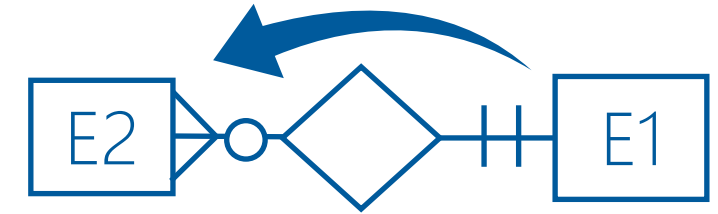
R1: Distributor(Name, Address, Phone, ...)

R2: Producer(Name, Studio, Address, Phone, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age)

42

Relationship: Multiple Many-One



R1: Distributor(Name, Address, Phone, ...)

R2: Producer(Name, Studio, Address, Phone, ...)

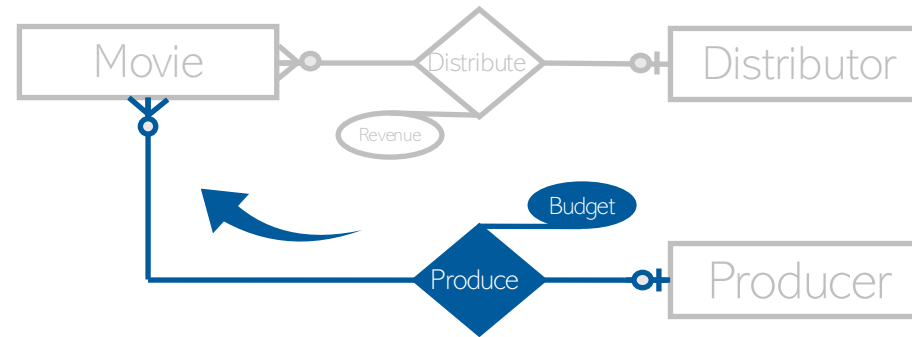
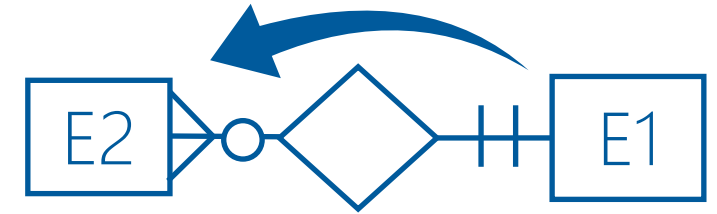
R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age,

Distributor.Name, Revenue)

FK₁: Optional (Nullable)

43

Relationship: Multiple Many-One



R1: Distributor(Name, Address, Phone, ...)

R2: Producer(Name, Studio, Address, Phone, ...)

R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age,

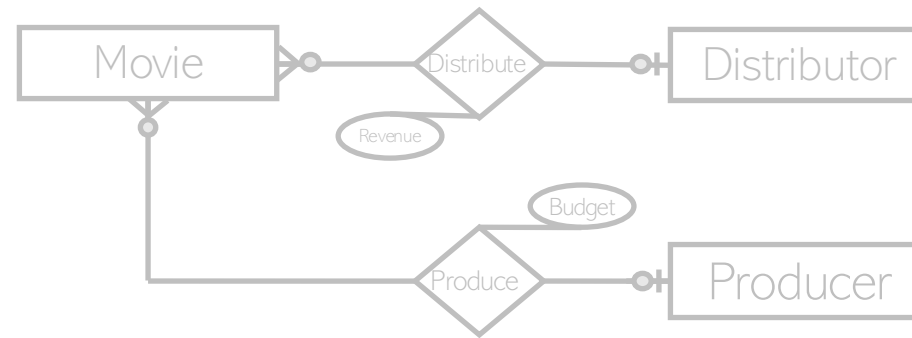
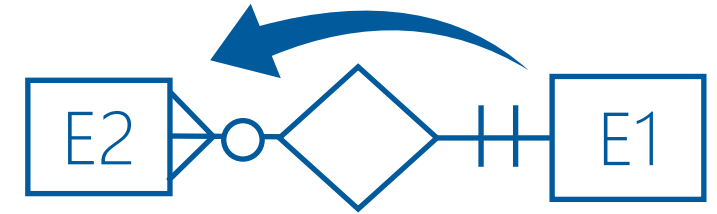
Distributor.Name, Revenue, Producer.Name, Budget)

FK₁: Optional (Nullable)

FK₂: Optional (Nullable)

44

Relationship: Multiple Many-One



R1: Distributor(Name, Address, Phone, ...)

R2: Producer(Name, Studio, Address, Phone, ...)

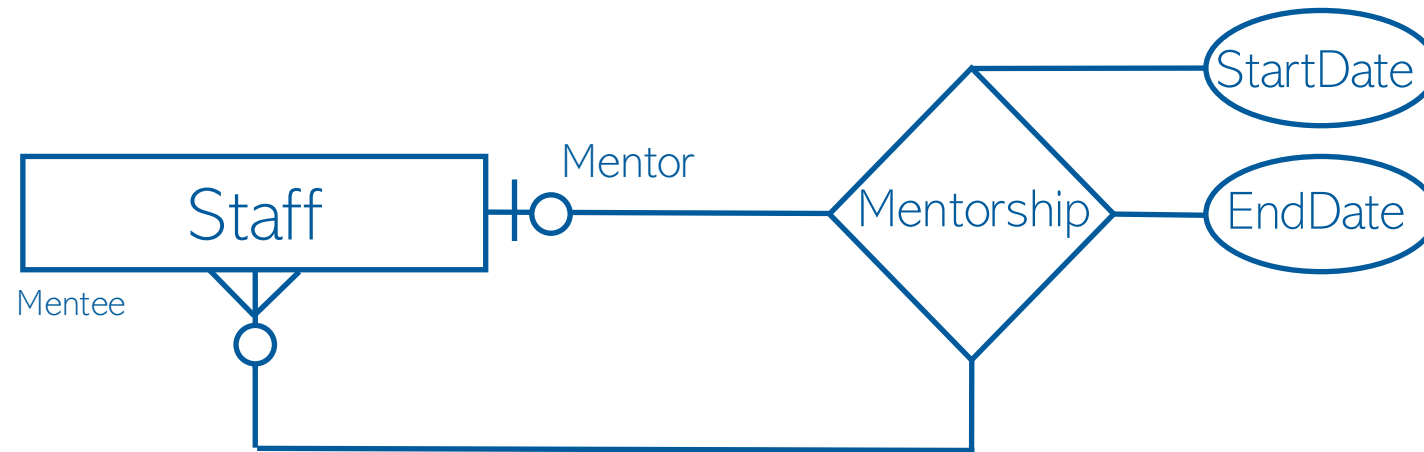
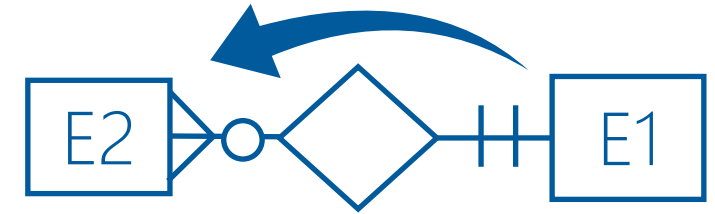
R3: Movie(Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime, Age, **Distributor.Name**, Revenue, **Producer.Name**, Budget)

FK₁: {Distributor.Name} Optional (Nullable)

FK₂: {Producer.Name} Optional (Nullable)

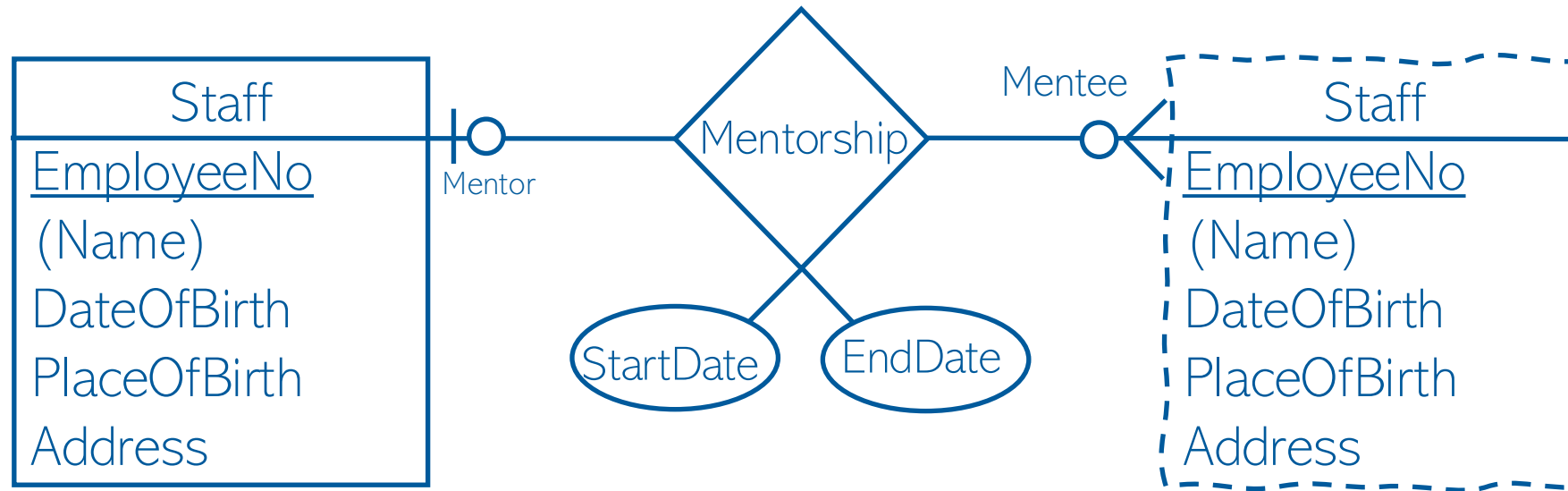
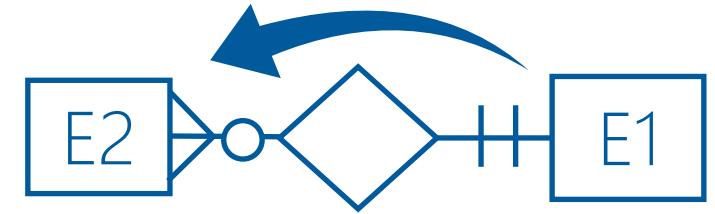
45

Relationship: Unary Many-One



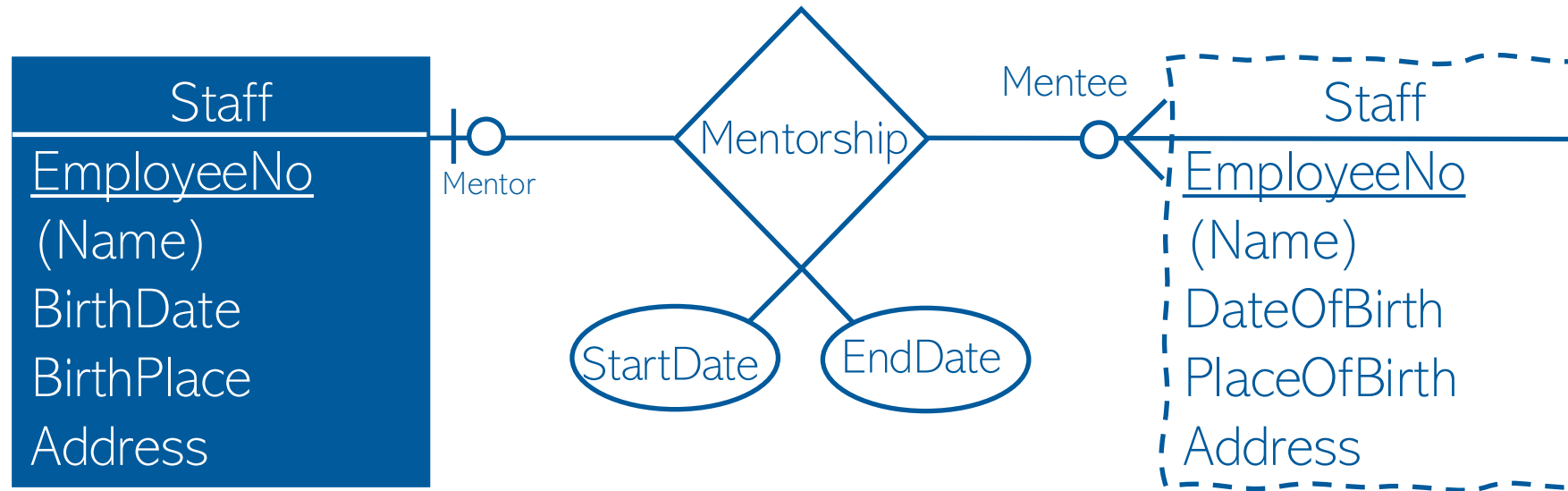
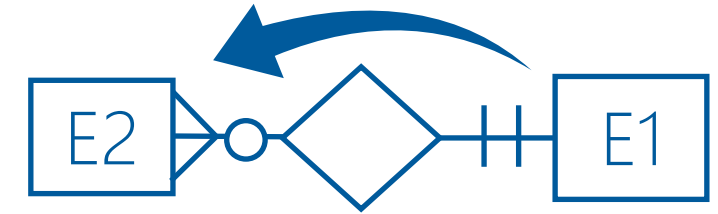
46

Relationship: Unary Many-One



47

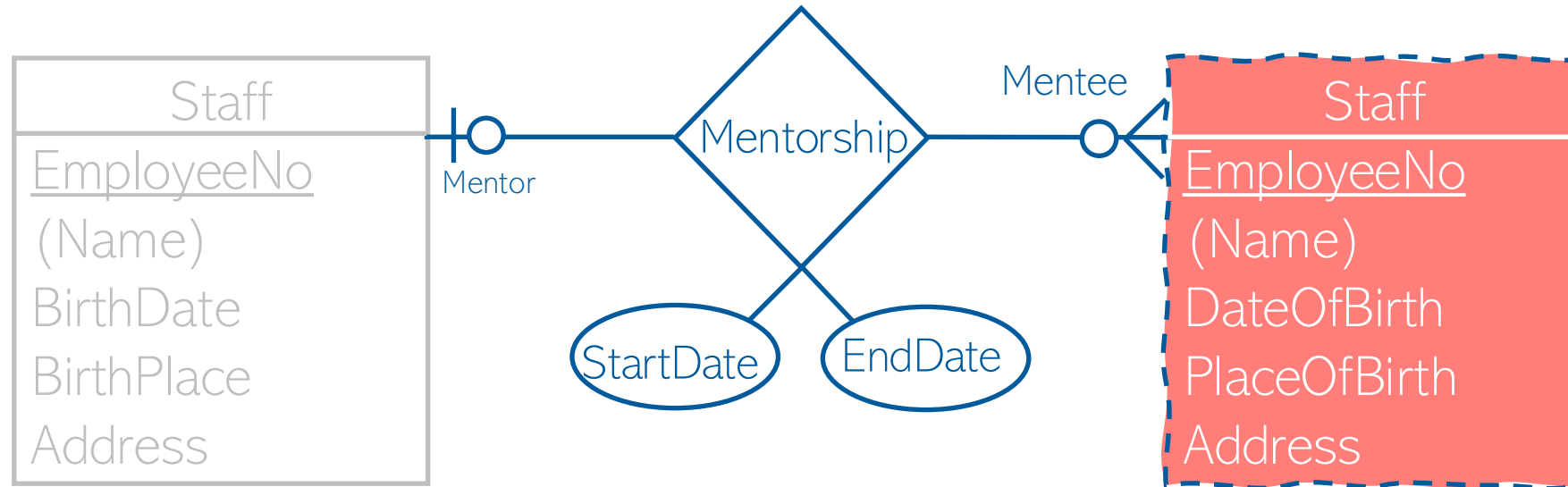
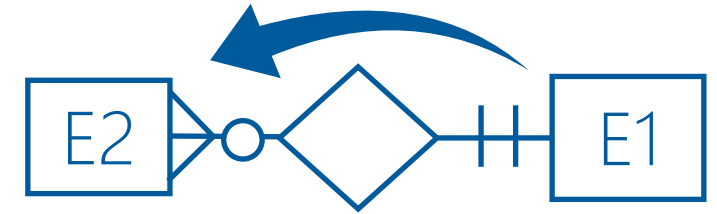
Relationship: Unary Many-One



R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

48

Relationship: Unary Many-One

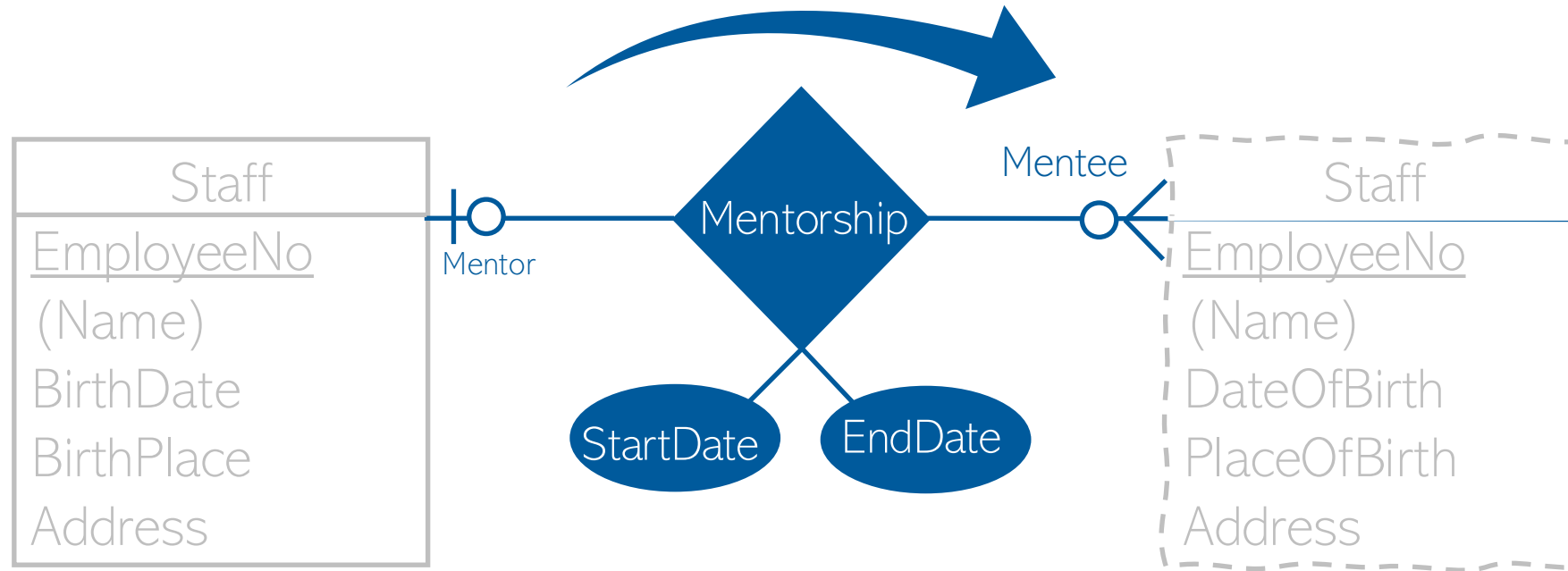
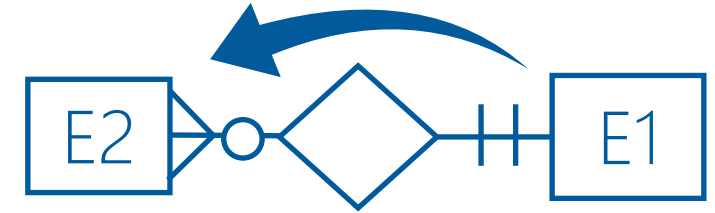


R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

49

Relationship: Unary Many-One



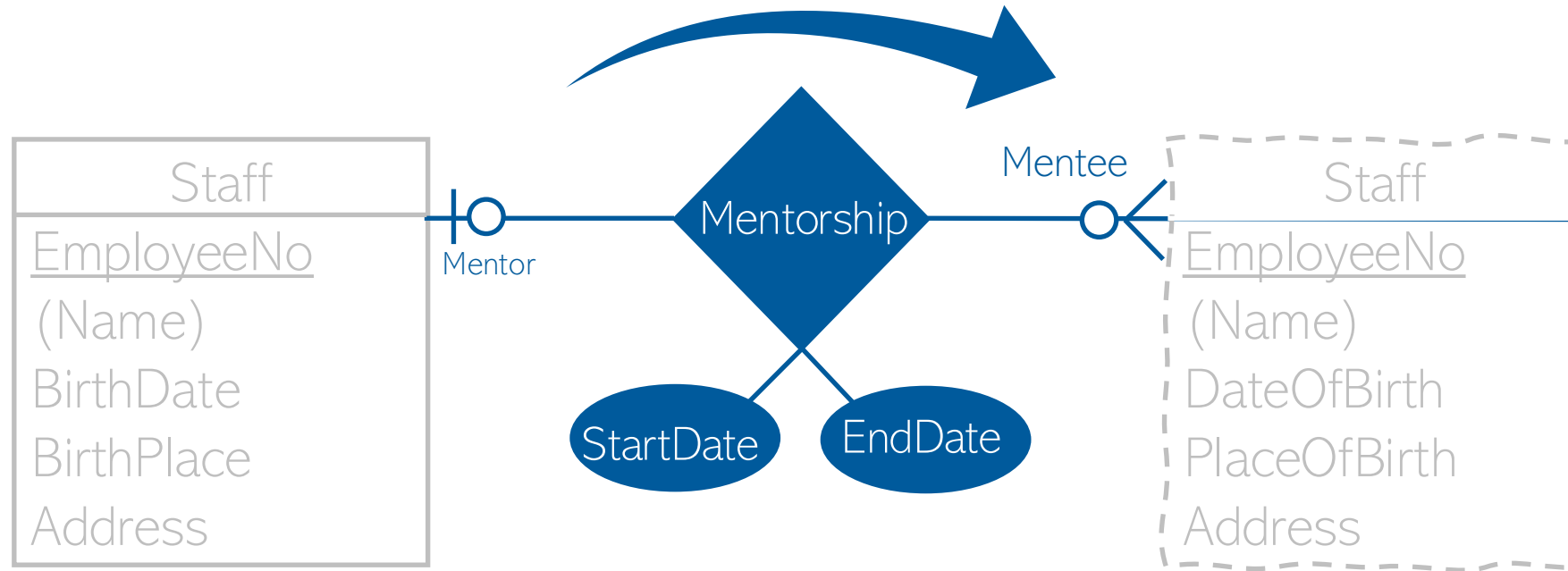
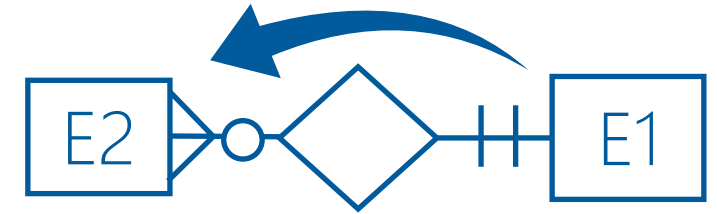
R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace,
Staff.EmployeeNo, StartDate, EndDate)

FK: Optional

50

Relationship: Unary Many-One



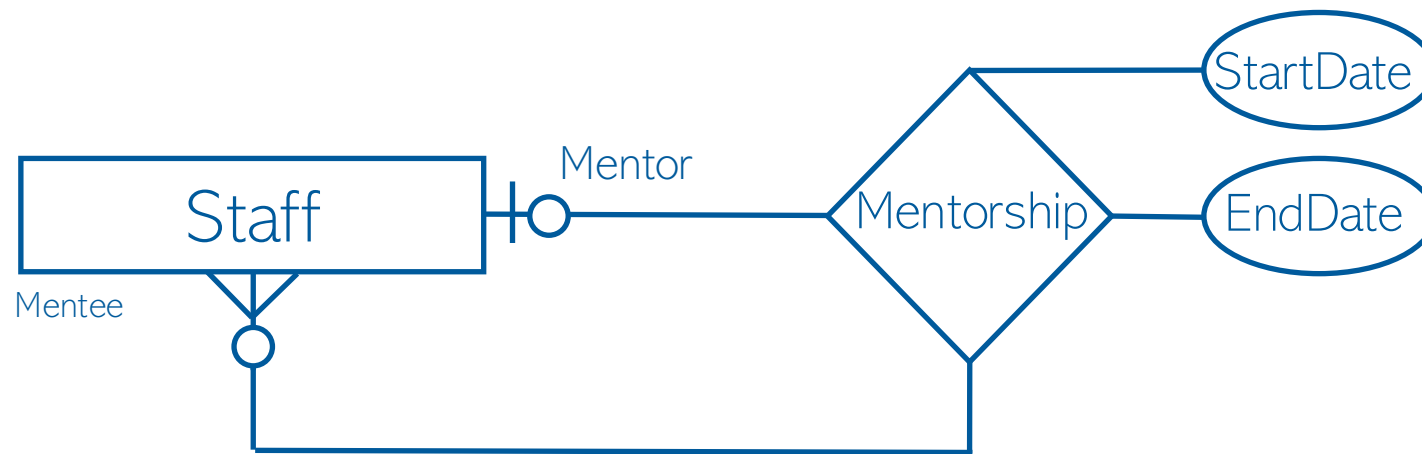
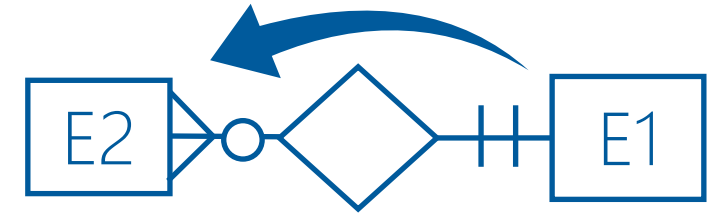
R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace)

R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace,
MentorNo, StartDate, EndDate)

FK: Optional; To avoid possible name conflict, we use role name.

51

Relationship: Unary Many-One

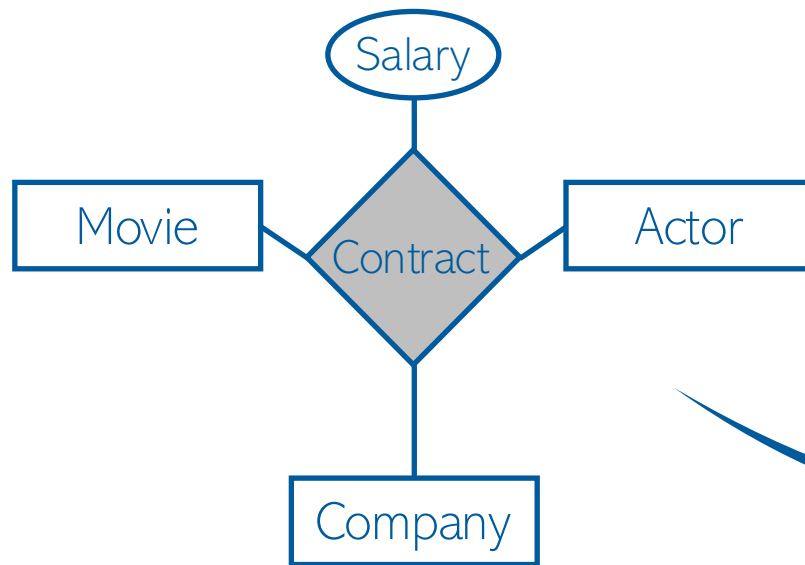
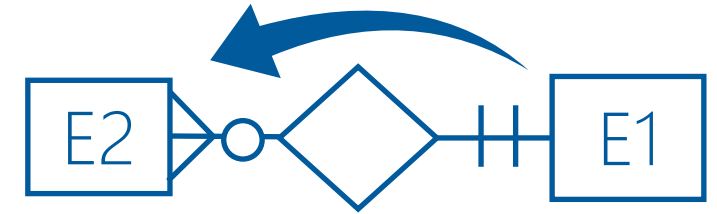


R1: Staff(EmployeeNo, Name, BirthDate, BirthPlace, MentorNo, StartDate, EndDate)

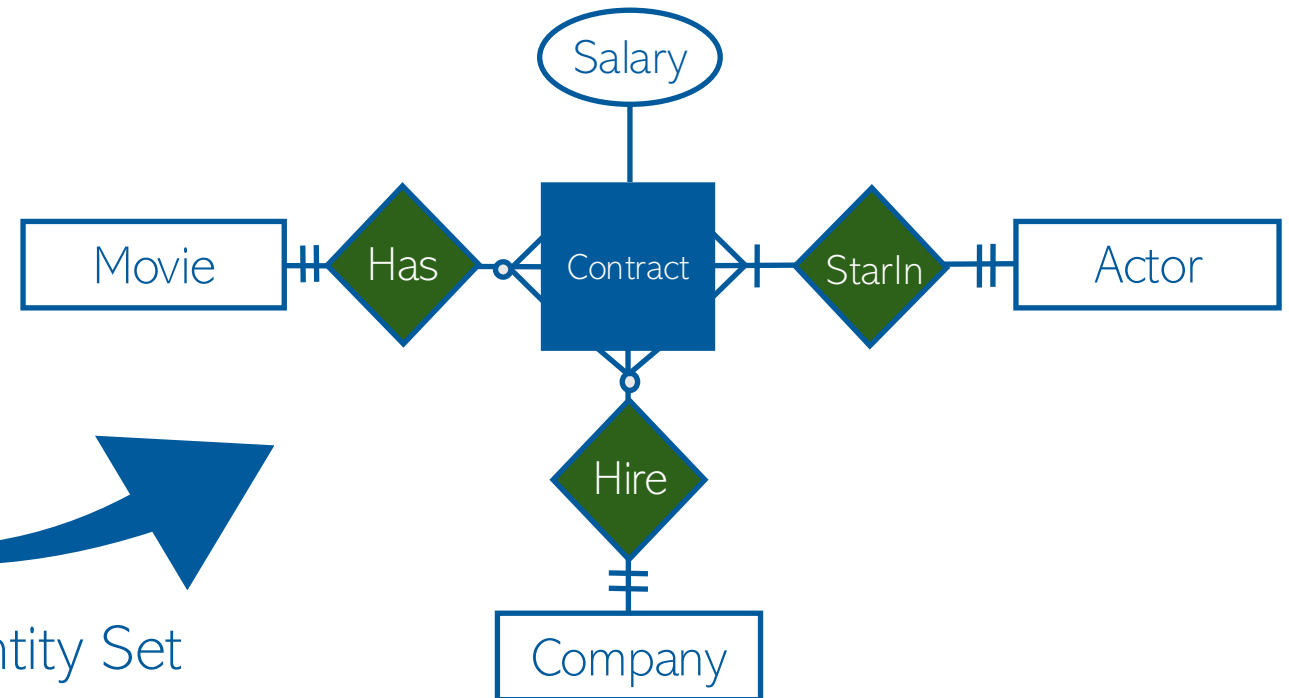
FK: {MentorNo} Optional

52

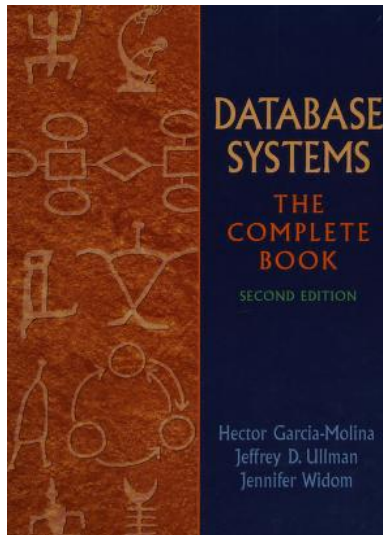
Relationship: Ternary Many-One



Connecting Entity Set



Outline: W04B



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation (Many-Many)

IIII) IS-A to Relation

Hands-On

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Outline: W04B

